A Non-Split Regge–Wheeler Self-Extension and a Defective Schwarzschild Resonance in Pure Weyl Gravity

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Abstract

We determine the differential-module and resonance structure of axial Schwarzschild perturbations in four-dimensional pure Weyl gravity. The six-state Bach system has an exact filtration with successive quotients given by two spin-two Regge–Wheeler modules and one spin-one Maxwell module. The repeated spin-two block is a non-split self-extension and is the first jet of a four-state spin-two/spin-one family.

After Liouville reduction, the extension is represented by a scalar projective cocycle modulo the symmetric-square operator

$$\mathcal{K}_U = D^3 + 4UD + 2D(U).$$

For axial $\ell = 2$ we prove the exact normal form

$$[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}\left[1 - \frac{2}{r}\right].$$

For every radiative multipole $\ell \geq 2$, the static class $[1 - 2/r]$ is nonzero in the rational cokernel of the corresponding static symmetric-square operator; the dipole is an explicit exact exception. Hence the axial $\ell = 2$ Bach extension has exact threshold valuation one: its original $1/\omega$ singularity is exact, but the surviving order-ω class cannot be removed by a gauge holomorphic at threshold. The class in brackets is exactly the transverse-traceless critical spin-two mass jet. Thus the local parameter relation is $m = (i\omega/2)\tau$, modulo rational triangular gauge. The required gauge grows linearly at infinity, precisely matching the derivative of the moving massive Jost phase. The long-range Coulomb tail cancels the apparent logarithmic phase derivative, so the gauge preserves the differentiated Jost classes modulo scalar normalization. The resulting generalized Schwarzschild QNM is the asymptotically flat quasinormal analogue of a critical-gravity logarithmic partner. Its canonical scalar mass-tangent class has a polynomial spacetime prefactor: relative to the ordinary QNM it is linear in a phase-adapted null time and in r. The fixed-frequency Jost derivative has no Coulomb $\log r$ term; the total derivative along the moving QNM branch is instead organized by the null coordinate.

At the covariant parent-operator level, the pure-Weyl metric Green function is exactly the first mass derivative of the Einstein–Weyl spin-two resolvent. This determines both leading Laurent coefficients and the isolated-resonance time contribution directly from the massive QNM velocity and projector derivative. The second mass derivative of the QNM frequency is likewise determined by the ordinary Regge–Wheeler finite resolvent part and augmented left/right resonant data. The quotient of the Bach shear by the scalar Evans function is a meromorphic spectral-velocity generating function: its QNM residues are the projective selectors, and its contour integral sums those selectors. The normalization-invariant first and second logarithmic spectral-flow forms have residues equal to the QNM velocity and acceleration; their weighted contour moments sum the corresponding spectral kinematics. The second critical jet separates velocity, acceleration, and projector motion in its three Laurent coefficients. At any simple QNM

the first mass velocity decides exactly between the defective and semisimple Weyl doublings. More generally, the order of contact of the massive QNM branch controls the maximal pole order of every higher critical mass jet.

At a simple spin-two quasinormal frequency the complete connection admits two possible local Smith types. A validated projective Evans calculation selects the defective type $(0, 0, 2)$ at one damped Schwarzschild mode. The geometric root vector is Einstein, whereas its generalized root vector has nonzero projection to the Ricci-carrier quotient. We further show that the compact-source, compact-observation outgoing Green operator on the Schwarzschild exterior has a nonzero rank-one pole of order two at this frequency. The same selector is the nonzero critical-mass velocity of the massive spin-two QNM. The resulting filtration-protected confluence sits inside an exact local two-parameter phase diagram. The physical mass axis has linear splitting, whereas a transverse filtration-breaking perturbation has square-root splitting. The spectral gap controls the projector norm, positive-metric condition number, resolvent crossover, and precision required to certify the physical slope. On the critical root fiber, the Jordan nilpotent uniquely forces a hyperbolic compatible nondegenerate Hermitian form up to chain equivalence and scale. The Einstein root is null, the full double-pole coefficient is its null rank-one square-zero operator, and a nondegenerate first-order branch-form limit occurs exactly for opposite Krein signatures. Finally, arbitrary analytic source, reconstruction, and observation maps preserve the leading pole precisely when their two resonant overlaps are nonzero. The resulting real isolated-resonance signal is a repeated-root damped sinusoid,

$$e^{-\gamma t}\big[(a_0 + a_1 t)\cos\Omega t + (b_0 + b_1 t)\sin\Omega t\big],$$

with a divided-difference continuation that remains regular when the two finite-mass resonances separate. The odd stress-energy source map is onto the smooth compact radial Regge–Wheeler forcings: every such forcing is realized by a conserved, traceless odd source. Choosing the forcing from the adjoint QNM therefore gives an exact nonzero source overlap. This is an existence theorem for a conformally admissible external source, not a calculation for a specified matter trajectory. For the exact axial reconstruction, the outgoing Einstein root has nonzero Bondi shear. The u-weighted double-pole term therefore has standard u/r strain falloff at future null infinity. By contrast, the time-independent generalized carrier component has $O(1)$ strain in the standard odd radiation gauge; it is not an ordinary asymptotically flat radiative mode. For the isolated critical kernel, a switched-on phase-locked drive has quadratic early-time buildup and a finite $1/(\gamma^2 + \Delta^2)$ steady amplitude. Phase-matched impulses show the corresponding finite-train N^2 law before damping resolves the sum, while a finite-window fixed-L^2-budget optimization selects the time-reversed conjugate kernel.

Promotion to a global physical resolvent and to a full retarded generalized ringdown expansion requires a separate analytic Fredholm realization, weighted endpoint domains, a specified conformal-matter source model, and control of the weakened generalized endpoint component.

1 Introduction and main results

Pure Weyl gravity has a fourth-order metric equation, but its Schwarzschild perturbations are not best understood as an unstructured fourth-order system. In the axial $\ell = 2$ sector the six-dimensional radial system has the filtered form

$$0 \subset M_2 \subset E_2 \subset M_{\text{Bach}}, \qquad \text{gr}\, M_{\text{Bach}} \simeq M_2 \oplus M_2 \oplus M_1,$$

where M_2 is the spin-two Regge–Wheeler module and M_1 is the spin-one Maxwell module. The middle object E_2 is a self-extension

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0.$$

The decisive point is that this extension is non-split over the rational differential field. The two spin-two layers therefore share scalar characteristics without defining two rationally separable radial branches.

This paper establishes ten central results.

1. The repeated spin-two block is a non-split Regge–Wheeler self-extension, realized exactly as a parameter jet.

2. Its projective extension class has the elementary normal form

$$[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}\left[1 - \frac{2}{r}\right].$$

   The static class in brackets is nonzero for every $\ell \geq 2$, with an explicit exact dipole control, so the $\ell = 2$ Bach threshold valuation is exactly one.

3. The class $[1 - 2/r]$ is exactly the transverse-traceless critical mass jet. Its linearly growing equivalence gauge is endpoint-compatible: the Coulomb tail cancels the apparent logarithmic derivative of the massive Jost phase.

4. The massive-QNM tangent is a quasinormal logarithmic partner in the critical-gravity sense. At fixed frequency its infinity Jost derivative has no Coulomb logarithm. Along the moving QNM branch its canonical spacetime class is $i\nu_n u_\sigma - \sigma ir/(2\omega_n) + O(1)$, where $u_\sigma = t + \sigma r_*$, and time translation acts by a length-two Jordan block.

5. At a certified damped Schwarzschild quasinormal frequency the connection is defective. The associated generalized root vector has a nonzero carrier quotient, and the exterior cut-off outgoing Green operator has a genuine rank-one double pole.

6. The selector equals the normalized critical-mass QNM velocity and is nonzero. A two-parameter unfolding proves that the exact Einstein filtration protects linear mass splitting, while reverse mixing produces generic square-root splitting. Its discriminant, projectors, positive metrics, local contour, and Krein form have exact gap-controlled limits. The critical nilpotent forces the compatible root-space form to be hyperbolic, excluding a positive self-adjoint Jordan realization.

7. The meromorphic function b_{B}/a generates infinitesimal spectral flow: its residue at each simple nonzero QNM is the selector, and its contour integral is the sum of the enclosed selectors. Vanishing first velocity is exactly the semisimple Smith branch, while the order of spectral contact controls the pole order of higher critical jets. The second logarithmic spectral-flow form has residue $\omega_n''(0)$ and exact weighted acceleration moments.

8. The covariant pure-Weyl metric Green function is the mass derivative of the massive spin-two resolvent. Its double coefficient is controlled by QNM motion, its simple coefficient by projector motion, and its local polynomial signal obeys an exact rank-one excitation/observation rule. The curvature $\omega_n''(0)$ is an exact augmented finite-part functional of ordinary Regge–Wheeler data. The second critical jet separates squared velocity, acceleration, and first and second projector motion in its three singular coefficients.

9. Analytic source and observation maps carry the leading pole to the measured channel by a rank-one overlap formula. A visible isolated reflected pair has the real repeated-root damped-sinusoid form displayed above. A divided-difference basis gives a uniform, finite-amplitude

continuation between the critical Jordan regime and two resolved simple modes. Exact odd-parity reconstruction proves that the Einstein root has nonzero Bondi shear and that the enhanced term has standard u/r strain, while the constant generalized carrier has weakened $O(1)$ strain. Every smooth compact radial forcing is realized by a conserved, traceless odd stress tensor, and an adjoint-matched choice has exact nonzero source overlap.

10. The isolated causal critical kernel has an exact coherent-forcing law. A phase-locked switched-on drive builds quadratically at early times and saturates with a squared Lorentzian denominator. Finite phase-matched pulse trains have an exact damped arithmetic–geometric sum, and the fixed-L^2-budget optimal finite-window drive is the time-reversed conjugate kernel.

The fifth result is stronger than a statement about a finite connection matrix. It is a meromorphic differential-response theorem for compact radial sources and local radial observations. It is deliberately weaker than a causal spacetime resolvent theorem. No global retarded $te^{i\omega_n t}$ ringdown assertion is made here; the polynomial term is proved only for the isolated local resonance contour.

Logical status

Result	Required input	Status
RW/RW/Maxwell filtration and partial jet	Exact rational identities and reconstruction rank	Exact
Non-split spin-two extension	Exhaustive rational coboundary obstruction	Exact for real $\omega > 0$
Cocycle and local critical-mass normal form	Exact substitution in $\mathbb{C}(r, \omega)$	Exact for $\omega \neq 0$
Static class and threshold valuation	Exact all-ℓ Laurent recurrence, static polynomial, and cubic obstruction	Exact
Endpoint-compatible mass comparison	Moving massive/Coulomb Jost germs and analytic normalization	Exact at the simple QNM
Quasinormal logarithmic partner	Massive-QNM tangent class, Coulomb cancellation, and Jordan chain	Exact reduced-mode theorem
Defective physical specialization	Winding-one contour, local selector, spin-one unit	Validated theorem
Non-Einstein generalized root	Defective Smith type and exact filtration	Exact consequence
Exterior cut-off outgoing Green double pole	Analytic Jost maps, transparent cuts, finite-interval reduction	Exact/validated theorem
Causal spacetime pole and generalized ringdown	Global weighted Fredholm and contour theory	Not claimed
Critical-mass unfolding and QNM velocity	Endpoint-compatible jet and nonzero certified selector	Exact/validated theorem
Spectral-velocity residues and contour sum	Analytic Evans domain, simple zeros, and endpoint-normalized mass derivative	Exact local/contour theorem
Contact-order pole hierarchy	Analytic moving simple pole and residue projector	Exact fixed-domain theorem
Spectral acceleration and second critical jet	Twice-differentiated Evans divisor and moving residue projector	Exact local/contour theorem
Canonical Krein–Jordan geometry	Nonzero length-two nilpotent and nondegenerate compatible Hermitian form	Exact finite-dimensional theorem
Observable transfer and detector normal form	Analytic source/reconstruction maps and isolated reflected QNM pair	Exact conditional reduced-mode theorem
Conserved traceless source nonannihilation	Odd stress projections, conservation identity, and adjoint-matched compact forcing	Exact existence theorem
Finite-time coherent forcing	Isolated critical kernel and declared coefficient-space L^2 budget	Exact conditional reduced-mode theorem
Two-parameter EP2 phase diagram	Exact Einstein filtration and a declared transverse mixing direction	Exact local theorem
Parent mass-derivative Green identity	Fixed-domain parent inverse and analytic massive QNM branch	Exact operator theorem
Second-order massive-QNM curvature	Ordinary finite resolvent part and augmented resonant pairing	Exact fixed-domain theorem
Local two-pole contour limit	Analytic normal form and residue calculus	Exact local theorem

2 Differential-module conventions

We work over

$$K = \mathbb{C}(r, \omega), \qquad D = f\partial_r, \qquad f = 1 - \frac{2}{r},$$

with $\omega \neq 0$ unless a threshold limit is stated explicitly. All rational gauges and module morphisms are over the differential field (K, D). We use $D(q)$, $D(U)$, and $D^3 q$, rather than primes, to avoid confusing the tortoise derivative with ∂_r.

The axial spin-two and spin-one scalar operators are

$$L_2 = D^2 + U_2, \qquad U_2 = \omega^2 - f\left(\frac{6}{r^2} - \frac{6}{r^3}\right),$$

$$L_1 = D^2 + U_1, \qquad U_1 = \omega^2 - f\frac{6}{r^2}.$$

The latter is the $\ell = 2$ Maxwell Regge–Wheeler equation. A rank-two scalar module is represented by its companion system on $(y, Dy)^T$.

The source Einstein layer is the kernel of the linearized Ricci/Schouten carrier map. The target spin-two quotient is its physical Regge–Wheeler carrier. The third quotient is the Maxwell class of the target Einstein gauge vector. Consequently, "non-Einstein" below means nonzero projection to the gauge-invariant carrier quotient, not failure of a coordinate gauge condition.

3 Exact axial RW/RW/Maxwell filtration

In the certified factor gauge the six-state system is

$$\begin{aligned} DX &= AX + EY + CZ, \\ DY &= AY + DZ, \\ DZ &= A_1 Z, \end{aligned} \tag{1}$$

where $X, Y, Z \in K^2$, the companion matrices A and A_1 represent L_2 and L_1, and the forcing blocks factor through a single scalar source:

$$E = USJ, \qquad C = USN.$$

The rational reconstruction has full row rank on the declared domain.

Theorem 3.1 (Filtered axial Bach module)**.** *For axial $\ell = 2$ Schwarzschild perturbations, the reduced six-state Bach module has a filtration whose successive simple factors are*

$$M_2, \qquad M_2, \qquad M_1.$$

The middle two-spin-two block is an extension E_2 of M_2 by M_2. The quotient M_1 is the Maxwell module of the carrier gauge vector. The filtration is exact but is not asserted to be a direct sum.

The theorem is an exact rational statement. It is independent of endpoint normalization, scattering conventions, and the later QNM specialization.

4 Partial-jet realization

Introduce the four-state family

$$\mathcal{B}(\tau) = \begin{pmatrix} A + \tau E & D + \tau C \\ 0 & A_1 \end{pmatrix}.$$

Let its fundamental matrix be

$$\Phi_4(\tau) = \begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix}.$$

Theorem 4.1 (Exact partial jet)**.** *The six-state generator in* (1) *is the first spin-two-row jet of* $\mathcal{B}(\tau)$ *at* $\tau = 0$*. Its fundamental matrix is*

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \qquad \det \Phi_6 = (\det P)^2 \det R.$$

Consequently any factor-adapted endpoint connection has the triangular form

$$T(\omega) = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix}, \qquad \det T = a^2 g.$$

Proof. Differentiate $D\Phi_4(\tau) = \mathcal{B}(\tau)\Phi_4(\tau)$ at $\tau = 0$. The equations for $P, \dot{P}, Q, \dot{Q}, R$ reproduce the three rows of (1) column by column. The determinant identity follows from block triangularity. The same identity applied to endpoint transition matrices gives the displayed connection form. □

At this stage τ is intrinsic. Section 15 proves its exact identification with the critical Einstein–Weyl mass jet and establishes compatibility with differentiated physical Jost germs.

5 Projective gauge and symmetric-square cohomology

Consider a scalar self-extension

$$Lx = (s_1 D + s_0)y, \qquad Ly = 0, \qquad L = D^2 + U. \tag{2}$$

Set

$$\mathcal{I} = s_0 - \frac{1}{2}D(s_1), \qquad \mathcal{K}_U = D^3 + 4UD + 2D(U).$$

Lemma 5.1 (Explicit triangular gauge)**.** *For* $q \in K$*, define*

$$Q(q) = qD - \frac{1}{2}D(q).$$

On every solution $Ly = 0$*,*

$$[L, Q(q)]y = -\frac{1}{2}\mathcal{K}_U(q)y.$$

Thus the change of lift $x \mapsto x + Q(q)y$ *changes the scalar extension density by*

$$\mathcal{I} \longmapsto \mathcal{I} - \frac{1}{2}\mathcal{K}_U(q).$$

Proof. Expand $LQ(q) - Q(q)L$, use the Leibniz rule for D, and then set $D^2 y = -Uy$. The coefficient of Dy cancels, while the coefficient of y is $-\frac{1}{2}[D^3 q + 4UDq + 2D(U)q]$. □

The extension is therefore represented by

$$[\mathcal{I}] \in K/\mathcal{K}_U K.$$

Because Lemma 5.1 contains the factor 1/2, removal of a term $\mathcal{K}_U q$ uses the triangular gauge $Q(2q)$. This factor does not alter the cohomology quotient but matters in an explicit frame calculation.

The equation $\mathcal{K}_U q = 0$ is the symmetric-square equation of L. If y_1, y_2 span the scalar solution space, then its solution space is spanned by $y_1^2, y_1 y_2, y_2^2$. The same operator controls both the projective obstruction and trace-free horizontal endomorphisms.

6 Mass-direction normal form of the Bach cocycle

For the axial $\ell = 2$ Bach extension, rational reduction gives

$$\mathcal{I}_{\text{Bach}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega}.$$

Theorem 6.1 (Mass-direction normal form)**.** *Let*

$$q(r,\omega) = -\frac{i}{120\omega}\left(15r + 13 + \frac{12}{r} + \frac{9}{r^2}\right).$$

For $\omega \neq 0$,

$$\mathcal{I}_{\text{Bach}} = \mathcal{K}_{U_2} q + \frac{i\omega}{2} f.$$

Consequently

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2}[f] \quad \text{in} \quad K/\mathcal{K}_{U_2}K.$$

In particular, $[f] \neq 0$ wherever the Bach self-extension is non-split.

Proof. Apply $D = f\partial_r$ three times to q and substitute

$$U_2 = \omega^2 - f\left(\frac{6}{r^2} - \frac{6}{r^3}\right)$$

in $\mathcal{K}_{U_2} q = D^3 q + 4U_2 Dq + 2D(U_2)q$. Clearing the common denominator $120r^4\omega$ reduces the claim to a polynomial identity in r and ω. Exact coefficient comparison gives the displayed formula. The quotient identity follows immediately. If $[f] = 0$, then $[\mathcal{I}_{\text{Bach}}] = 0$, which would split the extension. □

Theorem 6.2 (All-radiative-multipole static mass direction)**.** *Let*

$$\Lambda = \ell(\ell+1), \qquad U_{\ell,0} = -f\left(\frac{\Lambda}{r^2} - \frac{6}{r^3}\right), \qquad \ell \geq 2.$$

Then

$$\boxed{[f] \neq 0 \quad \text{in} \quad \mathbb{C}(r)/\mathcal{K}_{U_{\ell,0}}\mathbb{C}(r), \qquad f = 1 - \frac{2}{r}.}$$

Equivalently, no rational function p satisfies $\mathcal{K}_{U_{\ell,0}} p = f$.

Proof. A rational solution can have poles only at 0, 2, and infinity. At an ordinary finite point, a pole of order N would produce an uncancelled pole of order $N+3$, because the coefficient of ∂_r^3 is nonzero there.

Put $s = r - 2$. Since

$$D = \frac{s}{2+s}\partial_s = \frac{s}{2}\partial_s + O(s^2\partial_s), \qquad U_{\ell,0} = O(s),$$

a putative leading term $p \sim s^{-N}$, $N > 0$, gives

$$D^3 p = -\frac{N^3}{8}s^{-N} + O(s^{1-N}),$$

whereas the potential terms are $O(s^{1-N})$ and $f = O(s)$. Thus p has no pole at $r = 2$.

At $r = 0$, the indicial leading term is

$$\mathcal{K}_{U_{\ell,0}}(r^k) = -8(k-6)(k-2)(k+2)r^{k-6} + \cdots .$$

For $k < 0$, only $k = -2$ is exceptional. Write

$$p = a_{-2}r^{-2} + a_{-1}r^{-1} + a_0 + a_1 r + \cdots .$$

The first three Laurent equations give

$$\begin{aligned} a_{-1} &= \frac{\Lambda - 2}{3} a_{-2}, \\ a_0 &= \frac{(\Lambda-2)(5\Lambda-4)}{72} a_{-2}, \\ a_1 &= \frac{\Lambda(\Lambda-2)(\Lambda-1)}{72} a_{-2}. \end{aligned}$$

The first resonant compatibility condition is

$$\boxed{\frac{\Lambda^2(\Lambda-2)^2}{9} a_{-2} = 0.}$$

For $\ell \geq 2$, $\Lambda \geq 6$, so $a_{-2} = a_{-1} = a_0 = a_1 = 0$. In particular, the exceptional exponent does not produce an admissible pole. Hence a rational solution would be a polynomial.

If $p \sim a_M r^M$ at infinity, then

$$\mathcal{K}_{U_{\ell,0}} p = (M-1)(M-2\ell-2)(M+2\ell) a_M r^{M-3} + \cdots .$$

The static Regge–Wheeler equation has a polynomial solution

$$y_\ell(r) = \sum_{k=3}^{\ell+1} c_k r^k, \qquad c_k = \frac{(k-1)(k-2) - \Lambda}{2(k-3)(k+1)} c_{k-1} \quad (4 \leq k \leq \ell + 1).$$

Direct coefficient substitution proves $(D^2 + U_{\ell,0})y_\ell = 0$, so the symmetric-square identity gives

$$\mathcal{K}_{U_{\ell,0}}(y_\ell^2) = 0.$$

For $M > 3$, the leading coefficient above can vanish only at $M = 2\ell + 2 = \deg y_\ell^2$. Subtracting the appropriate multiple of y_ℓ^2 therefore leaves a polynomial of degree at most three:

$$p = a_0 + a_1 r + a_2 r^2 + a_3 r^3.$$

The preceding Laurent recurrence gives $a_0 = a_1 = 0$. Exact substitution then yields

$$(\Lambda + 4)a_2 = 15a_3, \qquad (3\Lambda + 4)a_2 = (6\Lambda + 33)a_3.$$

The inhomogeneous equation forces $a_3 \neq 0$. Eliminating a_2 would require

$$15(3\Lambda + 4) = (\Lambda + 4)(6\Lambda + 33), \qquad \text{or} \qquad \Lambda^2 + 2\Lambda + 12 = 0.$$

This has no real root and hence no value $\Lambda = \ell(\ell+1)$, $\ell \geq 2$. $\square$

Proposition 6.3 (Exceptional dipole control)**.** *For $\ell = 1$, $\Lambda = 2$, the class is instead exact:*

$$\boxed{q_1(r) = \frac{r^2}{6} + \frac{r^3}{15} + \frac{r^4}{36}, \qquad \mathcal{K}_{U_{1,0}} q_1 = f.}$$

Thus the rational static mass-direction obstruction distinguishes the radiative multipoles $\ell \geq 2$ from the dipole. This cohomological control is not by itself a complete physical analysis of the $\ell = 1$ gravitational sector.

Remark 6.4 (What remains for all-ℓ Bach nonsplitting)*.* Theorem 6.2 proves nontriviality of the mass direction, not the reduction of the Bach cocycle for every multipole. If a future calculation establishes

$$[\mathcal{I}_{\mathrm{Bach},\ell}] = c_\ell(\omega)[f], \qquad c_\ell(\omega) \neq 0,$$

then all-ℓ nonsplitting follows immediately. The remaining symbolic target is therefore the scalar coefficient $c_\ell(\omega)$; it is not claimed here.

Corollary 6.5 (Static exactness and exact threshold valuation)**.** *Write*

$$q(r,\omega) = \frac{q_{-1}(r)}{\omega}, \qquad q_{-1}(r) = -\frac{i}{120}\left(15r + 13 + \frac{12}{r} + \frac{9}{r^2}\right),$$

and $U_2(\omega) = U_{2,0} + \omega^2$. Then

$$\mathcal{K}_{U_2} = \mathcal{K}_{U_{2,0}} + 4\omega^2 D$$

and

$$\boxed{\mathcal{I}_{\mathrm{Bach}} = \frac{1}{\omega}\mathcal{K}_{U_{2,0}} q_{-1} + \omega\left(4Dq_{-1} + \frac{i}{2}f\right).}$$

In particular,

$$\operatorname*{Res}_{\omega=0} \mathcal{I}_{\mathrm{Bach}} = \mathcal{K}_{U_{2,0}} q_{-1}.$$

Thus the entire $1/\omega$ part is already exact for the static symmetric-square operator. In the meromorphically saturated rational projective lattice,

$$\boxed{\operatorname{ord}_{\omega=0}[\mathcal{I}_{\mathrm{Bach}}] = 1, \qquad \left.\frac{[\mathcal{I}_{\mathrm{Bach}}]}{\omega}\right|_{\omega=0} = \frac{i}{2}[f] \neq 0.}$$

Moreover, no rational triangular gauge holomorphic in ω can improve the representative $(i\omega/2)f$ to $O(\omega^2)$.

Proof. The decomposition follows by inserting $\mathcal{K}_{U_2} = \mathcal{K}_{U_{2,0}} + 4\omega^2 D$ in Theorem 6.1. The same theorem supplies the meromorphic representative $(i\omega/2)f$. Its coefficient at order ω is nonzero in the static cokernel by Theorem 6.2, proving exact valuation one.

If a holomorphic residual gauge removed that coefficient, its first relevant term $\omega q_1(r)$ would require

$$\mathcal{K}_{U_{2,0}} q_1 = \frac{i}{2} f$$

up to the fixed gauge-factor convention. This contradicts $[f] \neq 0$. $\square$

Remark 6.6 (Bulk class versus spectral frame). The original $1/\omega$ singularity is gauge-exact, while the surviving order-ω class is genuinely cohomological. The gauge producing the regular representative is itself $O(\omega^{-1})$ and grows at infinity, so it is singular in a holomorphic physical Jost frame. The remaining threshold problem is therefore entirely one of physical endpoint frames and Jost normalization. The corollary does not construct a threshold-uniform splitting, justify commuting the limits $\omega \to 0$ and $\tau \to 0$, or prove a physical estimate for $b(\omega)/a(\omega)^2$.

Remark 6.7 (Forward reference to the physical mass jet). At this stage the representative $f = 1-2/r$ is an intrinsic mass-shaped direction. Section 15 proves the exact identity

$$[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}[\mathcal{I}_{\mathrm{mass}}], \qquad m = \frac{i\omega}{2}\tau,$$

and establishes compatibility with differentiated physical Jost germs.

7 Period formula for the connection shear

Let y_1, y_2 be a scalar basis and set

$$Y = \begin{pmatrix} y_1 & y_2 \\ Dy_1 & Dy_2 \end{pmatrix}, \qquad W = y_1 Dy_2 - y_2 Dy_1.$$

Because the companion matrix has zero trace, $D(W) = 0$.

Theorem 7.1 (Symmetric-square period matrix). *For the family*

$$(D^2 + U + \tau V)y = 0,$$

the interaction-picture variation satisfies

$$Y^{-1}\partial_\tau A\, Y = \frac{V}{W}\begin{pmatrix} y_1 y_2 & y_2^2 \\ -y_1^2 & -y_1 y_2 \end{pmatrix}.$$

Consequently the spin-two connection shear is determined by the three periods

$$\int \frac{V}{W} y_1^2 \, dr_*, \qquad \int \frac{V}{W} y_1 y_2 \, dr_*, \qquad \int \frac{V}{W} y_2^2 \, dr_*.$$

Proof. In the basis $(y, Dy)^T$,

$$A(\tau) = \begin{pmatrix} 0 & 1 \\ -(U+\tau V) & 0 \end{pmatrix}, \qquad \partial_\tau A = \begin{pmatrix} 0 & 0 \\ -V & 0 \end{pmatrix}.$$

Insert

$$Y^{-1} = \frac{1}{W}\begin{pmatrix} Dy_2 & -y_2 \\ -Dy_1 & y_1 \end{pmatrix}$$

and multiply. Integration of the interaction-picture equation gives the three entries. $\square$

A coboundary $\mathcal{K}_U q$ changes these periods by endpoint terms. This is the connection-level form of Lemma 5.1: rational triangular gauges alter Jost frames but not the intrinsic divisor or Smith selector once the endpoint units are tracked.

8 Nonsplitting, endomorphisms, and Fitting ideals

Theorem 8.1 (Non-split physical-axis extension)**.** *For every real $\omega > 0$, the axial $\ell = 2$ spin-two block*

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0$$

is non-split over $\mathbb{C}(r)[D]$.

Certificate-supported exact proof. The rational splitting equation reduces, after exhaustive pole and degree bounds, to a finite coefficient system. Fixed minors of its coefficient and augmented matrices are

$$\det M_{[0,1,2]} = 3456\,\omega^{10}, \qquad \det[M \mid \mathrm{rhs}]_{[0,1,2,5]} = -645120\,i\,\omega^{9}.$$

They are simultaneously nonzero for every real $\omega > 0$, so the inhomogeneous coboundary equation is inconsistent. The fixed-minor argument also covers the former frame event $\omega^2 = 3$. □

The scalar Regge–Wheeler module is simple on the positive real axis and has scalar rational endomorphism ring. The only nonzero complex reducibility points allowed by the terminal Riccati ansatz with matching horizon and infinity signs are the algebraically special frequencies

$$\omega = \pm i \frac{(\ell - 1)\ell(\ell + 1)(\ell + 2)}{12}.$$

For $\ell = 2$, these are $\omega = \pm 2i$, away from the physical axis.

Theorem 8.2 (Local commutant)**.** *For real $\omega > 0$,*

$$\mathrm{End}_{\mathbb{C}(r)[D]}(E_2) \simeq \mathbb{C}[\varepsilon]/(\varepsilon^2).$$

Every endomorphism is $aI + bN$, where $N^2 = 0$. Hence the only idempotents are $0, I$, the only involutions are $\pm I$, and every diagonalizable local endomorphism is scalar.

Proof. The submodule M_2 is the unique simple submodule: a second simple submodule mapping nontrivially to the quotient would split the extension. An endomorphism therefore induces scalars a, d on the submodule and quotient. Naturality of the nonzero extension class gives $a = d$. Subtracting aI leaves a map that vanishes on the submodule and quotient, so it factors through the quotient back into the submodule and is a scalar multiple of the canonical nilpotent N. The corollaries follow from $(aI + bN)^k = a^k I + k a^{k-1} b N$. □

For the triangular connection

$$T = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix},$$

the Fitting ideals are

$$I_1 = (a, b, c, d, g), \qquad I_2 = (a^2, ad, bd - ac, ag, bg), \qquad I_3 = (a^2 g).$$

They give a normalization-independent local classification of all specializations.

9 Analytic Jost connections and Smith dichotomy

The moving Frobenius/Jost phases define analytic endpoint-selected planes on the certified spectral domains. Transport to a common ordinary radius and change to a factor-adapted frame. At a simple spin-two QNM ω_n, assume the spin-one coefficient $g(\omega_n) \neq 0$. Analytic row and column operations remove the spin-one row and column, leaving

$$T_2(\omega) = \begin{pmatrix} a(\omega) & b(\omega) \\ 0 & a(\omega) \end{pmatrix}, \qquad a(\omega_n) = 0, \quad a'(\omega_n) \neq 0.$$

Proposition 9.1 (Local Smith dichotomy). *At ω_n,*

$$\mathrm{Smith}(T) = \begin{cases} (0,0,2), & b(\omega_n) \neq 0, \\ (0,1,1), & b(\omega_n) = 0. \end{cases}$$

The first case has geometric multiplicity one and one length-two root chain; the second has two independent length-one roots.

Proof. Let $z = \omega - \omega_n$. Since $a = a_1 z + O(z^2)$ with $a_1 \neq 0$, the determinant valuation of the two-spin-two block is two. If $b(0) \neq 0$, its first Fitting ideal is the unit ideal, so its exponents are $(0,2)$. If $b(0) = 0$, all entries vanish to first order and the exponents are $(1,1)$. Appending the spin-one unit adds a zero exponent. □

Projectively, write the selected endpoint lines in a common chart as

$$v_H(\omega,\tau), \qquad v_\infty(\omega,\tau), \qquad \delta = v_\infty - v_H.$$

At a simple root,

$$\frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n,0)}{\delta_\omega(\omega_n,0)}.$$

This ratio is invariant under analytic nonzero endpoint units and under moving the matching section.

10 Certified QNM specialization

The validated contour is oriented positively and encloses the disk

$$\Omega_n = \left\{\omega : \left|\omega - \left(-0.3736716844180418 + 0.0889623156889357\, i\right)\right| \leq 10^{-7}\right\}.$$

Panelwise projective enclosures exclude zero on the contour, include all chart-transition units, and give winding number $+1$. Thus the disk contains exactly one simple spin-two zero.

On the localized disk, the intrinsic tangent is enclosed by

$$\delta_\tau(\Omega_n) \subset [0.0010932 \pm 7.60 \times 10^{-8}] + i[-0.0002195 \pm 8.89 \times 10^{-8}],$$

which excludes zero. An independent local calculation gives a spin-one mismatch modulus greater than 1/2500.

Theorem 10.1 (Certified defective Schwarzschild resonance). *The complete axial connection has Smith valuations*

$$(0,0,2)$$

at the unique QNM in Ω_n. The algebraic multiplicity of its spin-two divisor is two, its geometric multiplicity is one, and it has one length-two connection root chain.

Proof. The winding certificate gives one simple spin-two zero. The local tangent enclosure proves $b(\omega_n) \neq 0$, while the spin-one enclosure proves $g(\omega_n) \neq 0$. Proposition 9.1 applies. □

11 Resonant evaluation of the self-extension

The projective selector, adjoint overlap, root-chain coefficient, and Green-pole residue are different representations of one resonant functional. This is the finite-interval Evans/Melnikov mechanism; compare the general relation between Evans derivatives and adjoint generalized eigenfunctions in [7].

Let

$$L(\omega) : X \longrightarrow Y$$

denote the analytic finite-interval spin-two boundary pencil, with the endpoint conditions included either in its domain or as finite-dimensional components of an augmented range. At a simple resonance choose right and left states

$$L_n u_n = 0, \qquad L_n^* \widetilde{u}_n = 0, \qquad L_n = L(\omega_n),$$

and define, using the bilinear duality of the augmented boundary problem,

$$\alpha_n = \left\langle \widetilde{u}_n, L'(\omega_n) u_n \right\rangle, \qquad \alpha_n \neq 0.$$

Let the repeated block be

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix}, \qquad \beta_n = \langle \widetilde{u}_n, K(\omega_n) u_n \rangle .$$

All endpoint components of the augmented pencil and its duality are included in these pairings.

Theorem 11.1 (Resonant evaluation theorem)**.** *In endpoint frames compatible with the partial jet,*

$$\boxed{\kappa_n := \frac{b(\omega_n)}{a'(\omega_n)} = \frac{\beta_n}{\alpha_n} = -\omega_n'(0).}$$

Here $\omega_n(\tau)$ is the resonance branch of

$$L(\omega, \tau) = L(\omega) + \tau K(\omega)$$

through ω_n. The equality is invariant under a common analytic nonzero Evans unit.

Proof. Differentiate

$$L(\omega_n(\tau), \tau) u(\tau) = 0$$

at $\tau = 0$:

$$L_n \dot{u} + \omega_n'(0) L_n' u_n + K_n u_n = 0.$$

Pairing with $\widetilde{u}_n$ removes the first term and gives

$$\omega_n'(0) \alpha_n + \beta_n = 0.$$

On the connection side, differentiation of

$$a(\omega_n(\tau), \tau) = 0$$

gives

$$a'(\omega_n) \omega_n'(0) + \partial_\tau a(\omega_n, 0) = 0.$$

The exact partial-jet construction identifies $\partial_\tau a(\omega_n, 0) = b(\omega_n)$. Multiplication of the Evans coefficient by an analytic unit $u(\omega, \tau)$ multiplies both a' and b at the root by $u(\omega_n, 0)$, so the ratio is unchanged. □

Proposition 11.2 (Resonant functional on the extension class)**.** *The assignment*

$$\mathfrak{M}_n([K]) := \langle \widetilde{u}_n, K(\omega_n)u_n \rangle$$

is a well-defined linear functional on the local self-extension class. Thus nonsplitting is the statement $[K] \neq 0$*, while defectiveness at* ω_n *is the sharper statement* $\mathfrak{M}_n([K]) \neq 0$*.*

Proof. Under a triangular change of splitting,

$$K \longmapsto K + LQ - QL.$$

At the resonance,

$$\langle \widetilde{u}_n, (L_nQ - QL_n)u_n \rangle = \langle L_n^* \widetilde{u}_n, Qu_n \rangle - \langle \widetilde{u}_n, QL_nu_n \rangle = 0.$$

When endpoint conditions are encoded in an augmented pencil, the same calculation includes the corresponding boundary terms by definition. □

The theorem and proposition separate a global module property from its evaluation at one resonance. A nonzero extension class need not be defective at every QNM: the linear functional $\mathfrak{M}_n$ may vanish at isolated modes.

12 The generalized root vector is non-Einstein

Proposition 12.1 (Nonzero carrier in the generalized root)**.** *Assume*

$$a(\omega_n) = 0, \qquad a'(\omega_n) \neq 0, \qquad b(\omega_n) \neq 0.$$

The geometric root vector belongs to the source Einstein submodule, while every length-two root polynomial has first generalized coefficient with nonzero projection to the carrier quotient. In the normalized triangular frame that projection is

$$-\frac{a'(\omega_n)}{b(\omega_n)} = -\frac{\alpha_n}{\beta_n} = -\frac{1}{\kappa_n}.$$

Thus the defective doubled resonance is not two copies of an Einstein quasinormal mode: its generalized member is intrinsically non-Einstein.

Proof. Put $z = \omega - \omega_n$ and write

$$a = a_1 z + O(z^2), \qquad b = b_0 + O(z), \qquad a_1 b_0 \neq 0.$$

For

$$c(z) = c_0 + zc_1 + O(z^2), \qquad T_2(z)c(z) = O(z^2),$$

the order-zero equation gives $c_0 = e_X$, the source Einstein vector. Write $c_1 = xe_X + ye_Y$. The e_X component at order z is

$$a_1 + b_0 y = 0,$$

so $y = -a_1/b_0 \neq 0$. By Theorem 3.1, e_Y projects nontrivially to the Ricci-carrier quotient. Adding a multiple of e_X changes x, but cannot remove y. □

13 Exterior cut-off outgoing Green double pole

Fix ordinary radii $2 < r_H < r_I < \infty$ inside the certified analytic continuation domains. Let

$$H(\omega) : \mathbb{C}^3 \longrightarrow C^\infty([r_H, r_I]; \mathbb{C}^6)$$

be the analytic Poisson operator for horizon-selected homogeneous data, let $\Gamma_-(\omega)$ be the unwanted incoming trace at r_I, and set

$$T(\omega) = \Gamma_-(\omega)H(\omega).$$

For a compactly supported source F, let $P_H(\omega)F$ denote the inhomogeneous solution with zero horizon data. The outgoing solution is

$$R_{\text{out}}(\omega)F = P_H(\omega)F - H(\omega)T(\omega)^{-1}\Gamma_-(\omega)P_H(\omega)F. \tag{3}$$

Theorem 13.1 (Finite-interval outgoing Green pole)**.** *On a neighborhood of the certified frequency ω_n, the operator*

$$R_{\text{out}}(\omega) : C_c^\infty((r_H, r_I); \mathbb{C}^6) \longrightarrow C_{\text{loc}}^\infty((r_H, r_I); \mathbb{C}^6)$$

is meromorphic and has a nonzero rank-one pole of order two. Up to the endpoint trace sign convention, its principal coefficient is

$$\operatorname{Coeff}_{(\omega-\omega_n)^{-2}} R_{\text{out}} = \frac{b(\omega_n)}{a'(\omega_n)^2} H(\omega_n)e_X \otimes \left[e_Y^*\Gamma_-(\omega_n)P_H(\omega_n)\right].$$

Proof. All factors in (3) except T^{-1} are analytic. After analytic elimination of the spin-one unit, the singular block is

$$T_2^{-1} = \begin{pmatrix} a^{-1} & -ba^{-2} \\ 0 & a^{-1} \end{pmatrix},$$

and therefore

$$\operatorname{Coeff}_{z^{-2}} T_2^{-1} = -\frac{b(\omega_n)}{a'(\omega_n)^2} e_X \otimes e_Y^*.$$

The minus sign in (3) gives the displayed convention.

The range vector is nonzero because the homogeneous Poisson map is injective, so $H(\omega_n)e_X \neq 0$. The covector is nonzero by a cutoff Jost construction. Given any incoming coefficient q, choose the corresponding incoming Jost solution J_-q near r_I, multiply it by a smooth cutoff χ that vanishes near r_H and equals one near r_I, and set

$$F = (D - \mathbb{A}(\omega_n))(\chi J_- q).$$

Then F is compactly supported, $P_H(\omega_n)F = \chi J_- q$, and $\Gamma_- P_H(\omega_n)F = q$. Hence the incoming trace map from compact sources is onto. Taking $q = e_Y$ shows that the covector does not vanish. The tensor product is therefore nonzero and rank one. □

13.1 Exact transparent cuts

Put $x = r_*$. On the certified QNM neighborhood, let

$$u_H(x, \omega), \qquad u_\infty(x, \omega)$$

be analytic spin-two Jost solutions satisfying the future-horizon and outgoing-infinity conditions. The scalar Evans coefficient is, up to an analytic unit,

$$a(\omega) = W(u_H, u_\infty).$$

Let χ_{s} and χ_{o} be nontrivial smooth source and observation cutoffs with compact support. Choose finite cuts

$$x_- < \operatorname{supp}\chi_{\mathrm{s}} \cup \operatorname{supp}\chi_{\mathrm{o}} < x_+.$$

At the two cuts impose the exact transparent boundary conditions

$$W(u, u_H)|_{x_-} = 0, \qquad W(u, u_\infty)|_{x_+} = 0.$$

Their coefficients are analytic in ω. Uniqueness of the exterior Jost problems shows that the resulting finite-interval inverse agrees exactly, between the cutoffs, with the meromorphically continued exterior outgoing inverse.

Theorem 13.2 (Exterior cut-off Green pole)**.** *For any nontrivial compactly supported source and observation cutoffs,*

$$\chi_{\mathrm{o}} R_{\mathrm{ext}}(\omega)\chi_{\mathrm{s}}$$

has a genuine nonzero rank-one pole of order two at the certified Schwarzschild QNM. In the augmented Fredholm normalization its principal coefficient is

$$\boxed{\operatorname{Coeff}_{(\omega-\omega_n)^{-2}}\big(\chi_{\mathrm{o}} R_{\mathrm{ext}}\chi_{\mathrm{s}}\big) = -\frac{\beta_n}{\alpha_n^2}(\chi_{\mathrm{o}} u_n) \otimes (\widetilde{u}_n \chi_{\mathrm{s}}).}$$

Proof. The transparent conditions select exactly the restrictions of the exterior horizon and infinity Jost solutions. Hence

$$\chi_{\mathrm{o}} R_{\mathrm{ext}}(\omega)\chi_{\mathrm{s}} = \chi_{\mathrm{o}} R_{[x_-,x_+]}(\omega)\chi_{\mathrm{s}}.$$

Theorem 13.1 and (4) give the displayed principal coefficient. It is nonzero: $\beta_n \neq 0$, while a nonzero solution of a second-order ODE cannot vanish on a nonempty open interval. Thus $\chi_{\mathrm{o}} u_n$ and $\widetilde{u}_n \chi_{\mathrm{s}}$ are nonzero for every nontrivial smooth cutoff. □

Corollary 13.3 (Non-annihilation by certified reconstruction)**.** *The certified rational reconstruction into the axial metric/curvature variables does not annihilate the range $H(\omega_n)e_X$. Hence at least one reconstructed local observable has the same order-two radial response pole.*

In the finite-interval Fredholm normalization of Section 11, the scalar resolvent has

$$L(\omega)^{-1} = \frac{u_n \otimes \widetilde{u}_n}{\alpha_n(\omega - \omega_n)} + O(1).$$

The upper-right block of $\mathbb{L}^{-1}$ is $-L^{-1}KL^{-1}$, and Theorem 11.1 therefore gives the invariant principal coefficient

$$\boxed{\operatorname{Coeff}_{(\omega-\omega_n)^{-2}}(\mathbb{L}^{-1})_{12} = -\frac{\beta_n}{\alpha_n^2} u_n \otimes \widetilde{u}_n = -\frac{\kappa_n}{\alpha_n} u_n \otimes \widetilde{u}_n.} \tag{4}$$

The Poisson/trace coefficient in Theorem 13.1 and the exterior coefficient in Theorem 13.2 are the same rank-one residue expressed in endpoint and augmented Fredholm coordinates.

14 Conditional global Fredholm promotion

Theorem 13.2 is an exterior differential Green-operator theorem after compact source and observation localization. A global physical resolvent without cutoffs requires a separate closed-operator realization. The correct abstract block notation is

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

where $L(\omega) : D \subset X \to Y$ is an analytic Fredholm pencil of index zero and $K(\omega) : D \to Y$ is graph-norm bounded. Suppose

$$u_n \in \ker L(\omega_n), \qquad \widetilde{u}_n \in \ker L(\omega_n)^*,$$

and define using the Y^*-Y bilinear duality

$$\alpha_n = \langle \widetilde{u}_n, L'(\omega_n)u_n \rangle, \qquad \beta_n = \langle \widetilde{u}_n, K(\omega_n)u_n \rangle.$$

Proposition 14.1 (Conditional global pole formula)**.** *If L^{-1} is meromorphic near ω_n, its zero is simple, $\alpha_n \neq 0$, and $\beta_n \neq 0$, then*

$$\mathbb{L}^{-1} = \begin{pmatrix} L^{-1} & -L^{-1}KL^{-1} \\ 0 & L^{-1} \end{pmatrix}$$

has an order-two pole with principal coefficient

$$-\frac{\beta_n}{\alpha_n^2} u_n \otimes \widetilde{u}_n$$

in the upper-right block.

This proposition is standard block algebra once the Fredholm spaces and pairing exist. What remains open is not the exterior cut-off pole proved above, but the construction of:

1. closed weighted horizon and null-infinity domains;
2. a physical gauge quotient and bounded metric reconstruction;
3. a causal exterior resolvent on a Laplace contour;
4. a justified contour deformation and control of the non-pole part.

Only after these steps could the double pole be promoted to a rigorous $te^{i\omega_n t}$ contribution to retarded ringdown.

15 Critical Einstein–Weyl mass comparison

On the transverse-traceless spin-two sector, the quadratic Einstein–Weyl parent equation is

$$(\mathcal{E} + m\mathcal{A})\phi = 0,$$

where the parameter m has the dimensions of squared mass. Its Schwarzschild axial scalar reduction is

$$L_m(\omega)y = \left[D^2 + \omega^2 - V_2 - mf\right] y = 0, \qquad f = 1 - \frac{2}{r}. \tag{5}$$

For a differentiable family $y(m)$, differentiation at $m = 0$ gives

$$L_2\, \partial_m y = f\, y.$$

Theorem 15.1 (Exact local critical-mass-jet identification)**.** *The projective cocycle of the transverse-traceless critical mass jet is*

$$[\mathcal{I}_{\text{mass}}] = [f].$$

Consequently

$$\boxed{[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2}[\mathcal{I}_{\text{mass}}].}$$

With the convention $L_m = L_2 - mf$, the local first-order parameter relation is

$$\boxed{m = \frac{i\omega}{2}\tau}$$

modulo rational triangular gauge.

Proof. Equation (5) gives the extension equation $L_2\partial_m y = fy$, so its scalar projective density is f. Theorem 6.1 gives $[\mathcal{I}_{\text{Bach}}] = (i\omega/2)[f]$. Comparing the two tangent equations yields the parameter relation. □

This theorem closes the local transverse-traceless differential-module comparison. We next show that the same gauge matches the differentiated physical Jost classes.

15.1 Endpoint-compatible mass gauge

The rational representative in Theorem 6.1 satisfies

$$q(r,\omega) = -\frac{i}{8\omega}r + O(1), \qquad r \to \infty.$$

Proposition 15.2 (Forced moving-phase gauge)**.** *Every rational q satisfying*

$$\mathcal{I}_{\text{Bach}} = \mathcal{K}_{U_2}q + \frac{i\omega}{2}f$$

has the forced leading behavior

$$q(r,\omega) = -\frac{i}{8\omega}r + O(1).$$

Thus the equivalence gauge is necessarily unbounded in a fixed massless phase frame.

Proof. At infinity,

$$\mathcal{I}_{\text{Bach}} - \frac{i\omega}{2}f \longrightarrow -\frac{i\omega}{2}, \qquad U_2 \longrightarrow \omega^2.$$

A rational $q = o(r)$ is $O(1)$ and gives $\mathcal{K}_{U_2}q = o(1)$, so it cannot produce the nonzero constant. If $q \sim cr$, then $Dq \to c$, while $D^3q \to 0$ and $D(U_2)q \to 0$. Therefore

$$\mathcal{K}_{U_2}q \longrightarrow 4\omega^2 c.$$

Matching the constant gives $c = -i/(8\omega)$, uniquely. □

Factor normalization. The normalization in Lemma 5.1 must be retained. Define the field-redefinition operator

$$\boxed{Q_q = 2qD - D(q).}$$

Then

$$[L_2, Q_q]y = -\mathcal{K}_{U_2}(q)y, \qquad L_2 y = 0.$$

Consequently, if

$$L_2 x = \mathcal{I}_{\mathrm{Bach}} y, \qquad \widehat{x} = x + Q_q y,$$

then

$$\boxed{L_2 \widehat{x} = \frac{i\omega}{2} f y.} \tag{6}$$

The factor 2 in Q_q is essential.

Theorem 15.3 (Coulomb cancellation and differentiated Jost classes)**.** *Let $y_\sigma(m, r)$, $\sigma = \pm 1$, be a massive spin-two Jost germ at infinity, with $k = (\omega^2 - m)^{1/2}$ on the branch through $k(0) = \omega$. It has the form*

$$y_\sigma(m, r) = e^{\sigma i k r} r^{\rho_\sigma(m)} \left(1 + O(r^{-1})\right), \qquad \rho_\sigma(m) = \sigma i \left(2k + \frac{m}{k}\right).$$

In particular,

$$\rho'_\sigma(0) = 0, \qquad \frac{\partial_m y_\sigma(0, r)}{y_\sigma(0, r)} = -\frac{\sigma i}{2\omega} r + O(1).$$

Writing $c(\omega) = i\omega/2$, the rational gauge satisfies

$$\frac{Q_q y_\sigma}{y_\sigma} = \frac{\sigma r}{4} + O(1) = c(\omega) \frac{\partial_m y_\sigma(0, r)}{y_\sigma(0, r)} + O(1).$$

At the horizon the mass perturbation is multiplied by $f = O(r - 2)$; the indicial exponent is unchanged, q is regular, and Q_q preserves the phase-factored analytic ingoing germ. Hence, at either endpoint $e \in \{\mathcal{H}^+, \mathcal{I}^+\}$,

$$\boxed{\widehat{x}_e - c(\omega)\partial_m y_e(0) = \lambda_e(\omega) y_e}$$

for an analytic scalar λ_e.

Proof. At infinity the massive equation is

$$D^2 y + \left(k^2 + \frac{2m}{r} + O(r^{-2})\right) y = 0.$$

Substitution of the displayed ansatz at order r^{-1} gives

$$2\sigma i k \rho_\sigma + 4k^2 + 2m = 0,$$

and therefore the stated ρ_σ. Since

$$k'(0) = -\frac{1}{2\omega}, \qquad \frac{d}{dm}\left(2k + \frac{m}{k}\right)\bigg|_{m=0} = -\frac{1}{\omega} + \frac{1}{\omega} = 0,$$

the logarithmic derivative cancels. The remaining phase derivative gives the linear term in $\partial_m y_\sigma / y_\sigma$. Using $q = -ir/(8\omega) + O(1)$ and $D y_\sigma / y_\sigma = \sigma i \omega + O(r^{-1})$ gives

$$2q \frac{D y_\sigma}{y_\sigma} - D(q) = \frac{\sigma r}{4} + O(1).$$

Equation (6) shows that $\widehat{x} - c\,\partial_m y$ is homogeneous. The asymptotic calculation places it in the endpoint Jost line, proving the infinity statement. Regular Frobenius dependence and the unchanged indicial exponent prove the horizon statement. $\square$

15.2 Quasinormal logarithmic partner

Critical gravity on anti-de Sitter backgrounds is known to acquire generalized spin-two modes when the massive and massless operators coalesce. These are conventionally called logarithmic modes, and they arise as limits of the massive spin-two modes of the noncritical theory [1, 2]. The Schwarzschild construction above has the same mass-derivative and Jordan-chain mechanism, but its endpoint polynomial is adapted to asymptotically flat QNM conditions. Accordingly, we use the qualified term *quasinormal logarithmic partner*; *Jordan QNM* or *polynomial QNM partner* are equivalent descriptions of the reduced-mode result below.

Theorem 15.4 (Polynomial spacetime tangent of the quasinormal logarithmic partner)**.** *Let*

$$\Psi_m(t,r) = e^{i\omega_n(m)t} y_\sigma(\omega_n(m), m; r), \qquad \nu_n = \omega_n'(0) = \frac{2i}{\omega_n}\kappa_n,$$

where y_σ is a normalized massive infinity Jost germ and $\sigma = \pm 1$ labels its phase. Put

$$u_\sigma = t + \sigma r_*.$$

Then the normalization-independent tangent class

$$[d_m\Psi_m|_0] \in \mathcal{D}/\mathbb{C}\Psi_0$$

has the relative infinity asymptotics

$$\boxed{\frac{d_m\Psi_m|_0}{\Psi_0} = i\nu_n u_\sigma - \frac{\sigma i}{2\omega_n} r + O(1).} \tag{7}$$

In the Bach normalization $c(\omega_n) = i\omega_n/2$,

$$\boxed{\frac{c(\omega_n) d_m\Psi_m|_0}{\Psi_0} = -i\kappa_n u_\sigma + \frac{\sigma}{4} r + O(1).} \tag{8}$$

For the outgoing branch $\sigma = -1$,

$$u_\sigma = u = t - r_*, \qquad \frac{c(\omega_n) d_m\Psi_m|_0}{\Psi_0} = -i\kappa_n u - \frac{1}{4} r + O(1).$$

The $O(1)$ term changes under analytic normalization of the QNM family, but the displayed coefficients of u_σ and r do not. At fixed ω,

$$\partial_m \rho_\sigma(\omega, m)|_{m=0} = 0,$$

whereas along the moving QNM branch

$$\left.\frac{d}{dm}\rho_\sigma(\omega_n(m), m)\right|_{m=0} = 2\sigma i\nu_n.$$

Thus the fixed-frequency Jost tangent has no first-order Coulomb $\log r$ term; the total derivative is absorbed into the phase-adapted null coordinate in the full spacetime expression.

Proof. Fixed-frequency versus total mass derivatives. The fixed-frequency endpoint calculation of Theorem 15.3 gives

$$\partial_m \log y_\sigma(\omega, m; r)|_{m=0} = -\frac{\sigma i}{2\omega} r + O(1), \qquad \partial_m \rho_\sigma|_\omega = 0.$$

Along the QNM branch, however, both arguments of y_σ vary. Since the massless Coulomb exponent is $\rho_\sigma(\omega, 0) = 2\sigma i\omega$, the total radial derivative is

$$\frac{d}{dm} \log y_\sigma(\omega_n(m), m; r)\Big|_0 = \sigma i \left(\nu_n - \frac{1}{2\omega_n}\right) r + 2\sigma i\nu_n \log r + O(1).$$

Using $r_* = r + 2\log r + O(1)$ and differentiating the temporal phase gives

$$i\nu_n t + \sigma i \left(\nu_n - \frac{1}{2\omega_n}\right) r + 2\sigma i\nu_n \log r = i\nu_n(t + \sigma r_*) - \frac{\sigma i}{2\omega_n} r + O(1),$$

which proves (7). Multiplication by $c = i\omega_n/2$, followed by $\nu_n = 2i\kappa_n/\omega_n$, gives (8). Replacing Ψ_m by $v(m)\Psi_m$, with $v(0) \neq 0$, adds only $v'(0)\Psi_0/v(0)$, which changes the $O(1)$ representative but not the tangent class or either polynomial coefficient. □

Proposition 15.5 (Jordan time law). *Normalize a length-two root chain by*

$$HV_0 = \omega_n V_0, \qquad HV_1 = \omega_n V_1 + V_0.$$

Then

$$\boxed{e^{iHt} V_1 = e^{i\omega_n t}(V_1 + itV_0).}$$

The physical mass tangent is the same generalized class with the chain scale fixed by ν_n; hence its polynomial coefficient is $i\nu_n t$. The t-weighted spatial profile is the ordinary Einstein QNM V_0, while the time-independent generalized component V_1 has nonzero carrier quotient.

Proof. On the root space, $H = \omega_n I + N$ with $N^2 = 0$ and $NV_1 = V_0$. Therefore

$$e^{iHt} = e^{i\omega_n t}(I + itN).$$

□

Independent black-hole exceptional-point analyses likewise identify a linear-in-time contribution as the correct local resonant ansatz [3, 4]. Here that time law is tied additionally to the exact Bach self-extension, its mass tangent, its certified Smith type, and the nonzero exterior cut-off Green double pole.

Remark 15.6 (Not a literal Schwarzschild radial log mode). In the anti-de Sitter examples, differentiating a radial scaling exponent produces explicit logarithmic falloff. Here the two terms in

$$\rho_\sigma(m) = \sigma i \left(2k + \frac{m}{k}\right)$$

cancel at first order at fixed ω. The total QNM derivative does differentiate the massless Coulomb exponent because ω_n moves, but its logarithm combines with the temporal phase into $u_\sigma = t + \sigma r_*$. At fixed phase-adapted null time, the remaining endpoint polynomial is linear in r, not logarithmic. Thus the logarithmic terminology refers to the critical mass-derivative/Jordan ancestry, not to an independent $\log r$ falloff in the normalized Schwarzschild spacetime tangent.

Theorem 15.7 (Exact null-infinity reconstruction). *Let $E = 2\,\mathrm{EI2}$ be the unit outgoing Einstein line and let*

$$R = \mathrm{XI2} - \frac{i(16\omega^2 - 4i\omega - 5)}{\omega}\,\mathrm{XI3}$$

be the unit outgoing carrier line in the exact factor frame. After factoring their common retarded phase $e^{i\omega u}$, their Regge–Wheeler-gauge metric heads are

$$(H_0, H_1)_E = (-r,\, 2r) + O(1),$$
$$(H_0, H_1)_R = \left(\frac{3}{4}r^2 - \frac{3}{2}r,\, -\frac{3}{2}r^2\right) + O(1).$$

In particular,

$$(H_0, H_1)_R = -\frac{3}{4}r\,(H_0, H_1)_E + O(E).$$

Under the standard odd gauge transformation

$$h_u \mapsto h_u - i\omega\xi, \qquad h_2 \mapsto h_2 - 2\xi,$$

the radiation-gauge choice $\xi = h_u/(i\omega)$ gives

$$h_2 = \frac{2i}{\omega}h_u.$$

Writing $h_{AB} = h_2 X_{AB}$, the Bondi-shear map $\mathcal{O}_{\mathscr{I}^+}h = \lim_{r\to\infty} h_{AB}/r$ therefore satisfies

$$\boxed{\mathcal{O}_{\mathscr{I}^+}E = -\frac{2i}{\omega}X_{AB} \neq 0.} \tag{9}$$

The corresponding strains are

$$\frac{h^E_{AB}}{r^2} = -\frac{2i}{\omega r}e^{i\omega u}X_{AB} + O(r^{-2}),$$

and

$$\boxed{\frac{h^R_{AB}}{r^2} = \frac{3i}{2\omega}e^{i\omega u}X_{AB} + O(r^{-1}).} \tag{10}$$

Thus the linear scalar tangent does not cancel in the reconstructed time-independent generalized component: any root-chain representative with nonzero carrier quotient has weakened $O(1)$ strain. The odd trace-free tensor X_{AB} cannot be removed by a Weyl rescaling, and removing its leading term by an odd diffeomorphism would require a large boundary transformation rather than a standard small gauge change.

Proof. The displayed E head follows by substituting the exact EI2 series into the algebraic $C = 0$ reconstruction row. The R head follows by the displayed exact linear combination of the XI2 and XI3 metric heads. The odd gauge law and radiation-gauge reconstruction are the standard Schwarzschild formulas [5]; substituting the exact heads gives (9) and (10). Since the Einstein line is one power lower in r, adding an Einstein representative cannot cancel the carrier r^2 coefficient. □

Corollary 15.8 (Enhanced null-infinity coefficient)**.** *Normalize the resonant Einstein state by the line E. For the parent principal coefficient*

$$G_{-2} = -\frac{\nu_n}{4\alpha_{\mathrm{W}}\alpha_n}E \otimes \widetilde{u}_n,$$

the Bondi-shear principal coefficient is

$$\boxed{\mathcal{O}_{\mathscr{I}^+}G_{-2} = \frac{i\nu_n}{2\alpha_{\mathrm{W}}\alpha_n\omega_n}X_{AB} \otimes \widetilde{u}_n \neq 0.}$$

Consequently the isolated local contour contribution to strain from a source F contains

$$\boxed{-\frac{\nu_n}{2\alpha_{\mathrm{W}}\alpha_n\omega_n}\frac{u}{r}e^{i\omega_n u}X_{AB}\,\langle\widetilde{u}_n, S(\omega_n)F\rangle}$$

or, using $\nu_n = 2i\kappa_n/\omega_n$,

$$-\frac{i\kappa_n}{\alpha_{\mathrm{W}}\alpha_n\omega_n^2}\frac{u}{r}e^{i\omega_n u}X_{AB}\,\langle\widetilde{u}_n, S(\omega_n)F\rangle.$$

The spatial observation overlap is therefore nonzero. The coefficient for a specified physical source remains conditional on its adjoint overlap, and no global retarded contour deformation is asserted.

Theorem 15.9 (Conserved traceless source realization)**.** *Let $\ell \geq 2$, $\mu = (\ell-1)(\ell+2)$, and $\omega \neq 0$. In Schwarzschild coordinates, use the Martel–Poisson orientation $\epsilon_{tr} = 1$, hence $\epsilon^{tr} = -1$, and write the reduced odd master equation as*

$$\left(\partial_{r_*}^2 + \omega^2 - fV_{\mathrm{odd}}\right)\Psi = F, \qquad F \in C_c^\infty((2M,\infty)).$$

Define the odd stress projections

$$P_t = 0, \qquad P_r = \frac{\mu F}{2i\omega r f}, \qquad P = \frac{\partial_r(rF)}{2i\omega}.$$

Then

$$\nabla_a P^a + \frac{2}{r}r_a P^a - \frac{\mu}{r^2}P = 0$$

and the Cunningham–Price–Moncrief source is

$$S_{\mathrm{odd}} = -\frac{2r}{\mu}\epsilon^{ab}\nabla_a P_b = \frac{F}{f}.$$

Consequently the stress tensor

$$T^{aB} = \frac{P^a}{16\pi r^2}X^B, \qquad T^{AB} = \frac{P}{8\pi r^4}X^{AB},$$

with all other harmonic components zero, is smooth and radially compact, and satisfies

$$\boxed{\nabla_\mu T^{\mu\nu} = 0, \qquad T^\mu{}_\mu = 0.}$$

The complexified conserved-traceless odd source map is therefore onto the smooth compact reduced radial forcings.

Let $\widetilde{u}_n$ be the nonzero adjoint state at the certified QNM. Choose a nonnegative nonzero $\eta \in C_c^\infty((2M,\infty))$ supported where $\widetilde{u}_n \neq 0$, and set

$$F = \eta\,\overline{\widetilde{u}_n}.$$

Because the support is away from both endpoints, the augmented source pairing has no boundary term and

$$\boxed{\langle\widetilde{u}_n, F\rangle_{\mathrm{aug}} = \int \eta\,|\widetilde{u}_n|^2\,dr_* > 0.}$$

Thus a smooth compact conserved and traceless odd source excites the certified double pole. Together with Corollary 15.8, this source produces a nonzero isolated-contour $ue^{i\omega_n u}/r$ coefficient at future null infinity.

Proof. The harmonic norms are

$$\int X_A \overline{X^A}\, d\Omega = \ell(\ell+1), \qquad \int X_{AB} \overline{X^{AB}}\, d\Omega = \frac{1}{2}\mu\ell(\ell+1).$$

They show that the displayed T^{aB} and T^{AB} invert the standard odd source projections. Since $\sqrt{-\det g_{ab}} = 1$, one has

$$P^r = fP_r = \frac{\mu F}{2i\omega r}, \qquad \nabla_a P^a = \partial_r P^r.$$

Direct substitution gives both the conservation identity and $fS_{\rm odd} = F$. The remaining spacetime conservation equations vanish by the divergence-free odd vector harmonic, while tracelessness follows from $T^{ab} = 0$ and $\Omega_{AB} X^{AB} = 0$. The adjoint-overlap statement is immediate from the declared choice of F. □

Remark 15.10 (Endpoint domain and matter-source gate). Theorem 15.7 shows that the full time-independent generalized component is not contained in the standard fixed-boundary asymptotically flat radiative class. A global realization must either enlarge the endpoint domain to differentiated-Jost tangents or encode those tangents in an augmented boundary pencil. The u-weighted Einstein component is nevertheless an ordinary radiative field. On the source side, Theorem 15.9 proves nonannihilation for an explicit class of conformally admissible external sources, but not for a specified matter trajectory or an energy-condition-satisfying material model. In particular, the standard massive point-particle stress tensor has nonzero trace and cannot be coupled directly to the trace-free Bach equation without a conformal completion or compensator. Dynamical QNM excitation from generic Schwarzschild plunges supplies a useful methodology only after that matter model is fixed [6].

Theorem 15.11 (Critical-mass Evans derivative and QNM velocity)**.** *Let $a(\omega, m)$ be the massive spin-two Evans coefficient in analytic horizon and infinity Jost normalizations, and let $b_{\rm B}(\omega)$ be the Bach connection shear. There is an analytic scalar $h(\omega)$ such that*

$$\boxed{b_{\rm B}(\omega) = \frac{i\omega}{2}\,\partial_m a(\omega, 0) + h(\omega) a(\omega, 0).}$$

At the certified simple QNM,

$$b_{\rm B}(\omega_n) = \frac{i\omega_n}{2}\partial_m a(\omega_n, 0).$$

The massive QNM branch supplied by the implicit-function theorem therefore satisfies

$$\boxed{\left.\frac{d\omega_n}{dm}\right|_{m=0} = \frac{2i}{\omega_n}\kappa_n \neq 0.} \tag{11}$$

Proof. The endpoint corrections in Theorem 15.3 are scalar changes of the differentiated Jost normalizations. Their contribution to the differentiated connection coefficient is therefore an analytic multiple $h(\omega)a(\omega, 0)$. At a zero of a that term vanishes. Differentiating $a(\omega_n(m), m) = 0$ and using $\kappa_n = b_{\rm B}(\omega_n)/a'(\omega_n)$ gives

$$\kappa_n = -\frac{i\omega_n}{2}\left.\frac{d\omega_n}{dm}\right|_{m=0}.$$

The displayed formula follows, and it is nonzero because the certified selector satisfies $\kappa_n \neq 0$. □

Corollary 15.12 (Unified resonance invariant). *At the certified mode,*

$$\boxed{\kappa_n = \frac{b_n}{a_n'} = \frac{\beta_n}{\alpha_n} = -\frac{i\omega_n}{2}\,\frac{d\omega_n}{dm}\bigg|_0 .}$$

Consequently

$$(c_1)_Y = -\frac{1}{\kappa_n} = \frac{2}{i\omega_n\,\omega_n'(0)}$$

and, for the upper-right triangular Green block,

$$R_{-2}^{(12)} = -\frac{\kappa_n}{\alpha_n}u_n \otimes \widetilde{u}_n = \frac{i\omega_n}{2\alpha_n}\omega_n'(0)u_n \otimes \widetilde{u}_n.$$

Corollary 15.13 (Reflected EP2 pair). *With the $e^{i\omega t}$ convention, the real Schwarzschild coefficients and QNM endpoint conditions obey*

$$L(-\overline{\omega}, m) = \overline{L(\omega, m)}.$$

Hence the certified defective resonance near

$$\omega_n = -0.3736716844180418 + 0.0889623156889357\,i$$

has a reflected defective partner near

$$\boxed{\omega_n^\sharp = +0.3736716844180418 + 0.0889623156889357\,i.}$$

Their velocities and projective selectors satisfy

$$\nu_n^\sharp = -\overline{\nu_n}, \qquad \kappa_n^\sharp = -\overline{\kappa_n}.$$

In particular, nonvanishing of the certified selector supplies a symmetry-related pair of EP2s. If P_n, C_{-2}, and C_{-1} denote respectively the ordinary residue and the two singular coefficients of the critical response, then

$$P_n^\sharp = -\overline{P_n}, \qquad C_{-2}^\sharp = \overline{C_{-2}}, \qquad C_{-1}^\sharp = -\overline{C_{-1}}.$$

The even/odd signs follow from the pole order under $\omega \mapsto -\overline{\omega}$, and allow the pair to combine into a real perturbation.

15.3 Spectral-velocity generating function

The endpoint-normalized mass comparison also packages the selectors of many QNMs into one meromorphic function.

Theorem 15.14 (Spectral-velocity residues and contour sum). *Let Ω be a domain on which $a(\omega, m)$, $b_{\mathrm{B}}(\omega)$, and the endpoint Jost normalizations in Theorem 15.11 are analytic. Away from the zeros of $a(\omega, 0)$, define*

$$\mathscr{S}(\omega) = \frac{b_{\mathrm{B}}(\omega)}{a(\omega, 0)}.$$

Then

$$\boxed{\mathscr{S}(\omega) = \frac{i\omega}{2}\,\partial_m \log a(\omega, 0) + h(\omega).}$$

If $\omega_j \neq 0$ is a simple zero of $a(\omega, 0)$, then

$$\boxed{\operatorname*{Res}_{\omega=\omega_j} \mathscr{S}(\omega) = \kappa_j = -\frac{i\omega_j}{2}\omega_j'(0).}$$

Consequently, if a positively oriented contour $\Gamma \Subset \Omega$ contains only simple zeros of a and no zero lies on Γ, then

$$\boxed{\frac{1}{2\pi i}\oint_\Gamma \frac{b_{\mathrm{B}}(\omega)}{a(\omega,0)}\,d\omega = \sum_{\omega_j \in \operatorname{int}\Gamma} \kappa_j.} \tag{12}$$

Equivalently,

$$\boxed{\sum_{\omega_j \in \operatorname{int}\Gamma} \omega_j\omega_j'(0) = 2i\,\frac{1}{2\pi i}\oint_\Gamma \frac{b_{\mathrm{B}}}{a}\,d\omega.}$$

Proof. Divide the identity in Theorem 15.11 by $a(\omega, 0)$. Near a simple moving zero,

$$a(\omega, m) = u(\omega, m)\big(\omega - \omega_j(m)\big), \qquad u(\omega_j, 0) \neq 0,$$

and therefore

$$\partial_m \log a(\omega, 0) = -\frac{\omega_j'(0)}{\omega - \omega_j} + \text{holomorphic}.$$

Multiplication by $i\omega/2$ gives residue $-i\omega_j\omega_j'(0)/2 = \kappa_j$. The analytic normalization term h has zero contour integral, so the residue theorem proves (12); the weighted form follows from $\omega_j\omega_j'(0) = 2i\kappa_j$. $\square$

Corollary 15.15 (Reflection audit)**.** *If Γ and its enclosed zero set are invariant under $\omega \mapsto -\overline{\omega}$, then the selector sum in (12) is purely imaginary. Indeed, a reflected pair contributes*

$$\kappa_j + \kappa_j^\sharp = \kappa_j - \overline{\kappa_j} = 2i\,\operatorname{Im}\kappa_j.$$

A fixed zero on the imaginary axis has purely imaginary selector by the same symmetry. This is an exact audit for a future validated multi-QNM contour calculation; no such calculation is claimed here.

Proposition 15.16 (Complete first-jet dichotomy at a simple QNM)**.** *Let $\omega_j \neq 0$ be a simple scalar QNM and assume the spin-one factor is a local unit. Then exactly one of the following alternatives holds:*

$$\omega_j'(0) \neq 0 \Longleftrightarrow b_{\mathrm{B}}(\omega_j) \neq 0 \Longleftrightarrow \operatorname{Smith}(T) = (0,0,2),$$
$$\omega_j'(0) = 0 \Longleftrightarrow b_{\mathrm{B}}(\omega_j) = 0 \Longleftrightarrow \operatorname{Smith}(T) = (0,1,1).$$

In the first case the first critical jet has a double pole and a length-two generalized root. In the second case it has two semisimple length-one roots and no double pole. Its projected critical response can nevertheless retain the simple coefficient

$$C_{-1} = -\dot{P}_j.$$

Thus the semisimple case splits further into a shape-sensitive alternative $\dot{P}_j \neq 0$, with a simple pole, and a first-jet-invisible alternative $\dot{P}_j = 0$, in which the projected first mass derivative is holomorphic at ω_j.

Proof. At a nonzero simple QNM, Theorem 15.11 identifies $b_{\mathrm{B}}(\omega_j)$ with $(i\omega_j/2)\partial_m a(\omega_j, 0)$, while implicit differentiation gives $\omega_j'(0) = -\partial_m a/\partial_\omega a$. Proposition 9.1 gives the two Smith types. Finally, Proposition 16.2 gives $C_{-2} = -\omega_j'(0)P_j$ and $C_{-1} = -\dot{P}_j$. □

The critical determinant alone cannot make this distinction: $\det T_2 = a^2$ has algebraic multiplicity two in both Smith branches. The shear b_{B}, equivalently the transverse mass derivative of the Evans divisor, supplies the missing unfolding information.

15.4 Boundary transgression as an audit

For finite cuts, the direct field-redefinition convention gives

$$\boxed{\beta^{\mathrm{Bach}}_{n,[x_-,x_+]} - \frac{i\omega_n}{2}\beta^{\mathrm{mass}}_{n,[x_-,x_+]} = -\left[W(\widetilde{u}_n, Q_q u_n)\right]_{x_-}^{x_+}.}$$

In the augmented Jost pencil, the endpoint-map derivatives absorb this term. Theorem 15.11 shows that an explicit evaluation of the individual endpoint constants is unnecessary at the zero divisor: their net effect is the term $h(\omega)a(\omega, 0)$.

16 Universal critical resonance and the covariant parent

Theorem 16.1 (Universal critical-resonance criterion)**.** *Let $L(\omega) : D \to Y$ be an analytic Fredholm pencil with fixed domain, and suppose ω_n is a simple resonance. Let*

$$L_n u_n = 0, \qquad L_n^* \widetilde{u}_n = 0, \qquad \alpha_n = \langle \widetilde{u}_n, L_n' u_n \rangle \neq 0.$$

For an analytic deformation $L_m = L + mA$, put

$$\beta_n = \langle \widetilde{u}_n, A(\omega_n) u_n \rangle$$

in the augmented endpoint pairing. If $R = L^{-1}$, then

$$C(\omega) = R(\omega)A(\omega)R(\omega)$$

has expansion

$$\boxed{C(\omega) = \frac{\beta_n}{\alpha_n^2}\frac{u_n \otimes \widetilde{u}_n}{(\omega-\omega_n)^2} + \frac{C_{-1}}{\omega-\omega_n} + C_{\mathrm{hol}}(\omega).}$$

Consequently C has an exact pole of order two if and only if $\beta_n \neq 0$; at a simple resonance it cannot have a pole of order three. If $\omega_n(m)$ is the deformed resonance branch, then

$$\boxed{\nu_n := \omega_n'(0) = -\frac{\beta_n}{\alpha_n}, \qquad C_{-2} = -\nu_n P_n, \qquad P_n = \frac{u_n \otimes \widetilde{u}_n}{\alpha_n}.}$$

Proof. Write

$$R(\omega) = \frac{P_n}{\omega - \omega_n} + H_n + O(\omega - \omega_n).$$

The only second-order term in RAR is

$$P_n A_n P_n = \frac{\beta_n}{\alpha_n^2} u_n \otimes \widetilde{u}_n.$$

Differentiating $L_m(\omega_n(m))u_n(m) = 0$ and pairing with $\widetilde{u}_n$ gives $\beta_n + \nu_n \alpha_n = 0$. □

Proposition 16.2 (Canonical simple pole and tangent state)**.** *Let $z = \omega - \omega_n$ and expand*

$$R(\omega) = \frac{P_n}{z} + H_n + O(z), \qquad A(\omega) = A_n + zA'_n + O(z^2).$$

Then the critical response $C = RAR = -\partial_m R_m|_{m=0}$ has

$$\boxed{C_{-2} = P_n A_n P_n = -\nu_n P_n}$$

and

$$\boxed{C_{-1} = P_n A_n H_n + H_n A_n P_n + P_n A'_n P_n = -\dot{P}_n.}$$

The term $P_n A'_n P_n$ is absent only when the deformation operator is frequency independent in the fixed-domain pencil.

Let

$$g_n = (A_n + \nu_n L'_n)u_n.$$

Then the right and left tangent classes are

$$\boxed{[\dot{u}_n] = [-H_n g_n] \in D/\mathbb{C}u_n, \qquad [\dot{\widetilde{u}}_n] = [-\widetilde{u}_n(A_n + \nu_n L'_n)H_n] \bmod \mathbb{C}\widetilde{u}_n.}$$

Proof. Direct multiplication of the two displayed expansions gives both Laurent coefficients, including the A'_n term. Differentiating the moving simple pole gives $C_{-1} = -\dot{P}_n$.

The differentiated QNM equation is

$$L_n \dot{u}_n + g_n = 0, \qquad \langle \widetilde{u}_n, g_n \rangle = \beta_n + \nu_n \alpha_n = 0.$$

The Laurent identities imply

$$L_n H_n = I - L'_n P_n.$$

Since $P_n g_n = 0$, the vector $-H_n g_n$ solves the equation; the remaining freedom is a multiple of u_n. The left formula follows from the corresponding right Laurent identity. □

Theorem 16.3 (Second-order massive-QNM curvature)**.** *Let*

$$L_m(\omega) = L(\omega) + mA(\omega), \qquad \omega_n(m) = \omega_n + \nu_n m + \frac{1}{2}\xi_n m^2 + O(m^3),$$

be a fixed-domain analytic Fredholm realization of the massive QNM branch, with all endpoint-map derivatives included in the augmented operator. Put

$$B_n = A_n + \nu_n L'_n, \qquad \alpha_n = \langle \widetilde{u}_n, L'_n u_n \rangle.$$

Then

$$\boxed{\xi_n = \omega''_n(0) = \frac{2\langle \widetilde{u}_n, B_n H_n B_n u_n \rangle - \nu_n^2 \langle \widetilde{u}_n, L''_n u_n \rangle - 2\nu_n \langle \widetilde{u}_n, A'_n u_n \rangle}{\alpha_n}.}$$

The formula is independent of the analytic normalization chosen for $u_n(m)$.

Proof. The first differentiated equation and the finite-part reconstruction are

$$L_n \dot{u}_n + B_n u_n = 0, \qquad \dot{u}_n = -H_n B_n u_n + c\, u_n.$$

Differentiating the moving QNM equation a second time gives

$$L_n\ddot{u}_n + 2B_n\dot{u}_n + \left(\xi_n L_n' + \nu_n^2 L_n'' + 2\nu_n A_n'\right) u_n = 0.$$

Pairing with $\widetilde{u}_n$, substituting the finite-part formula, and using

$$\langle \widetilde{u}_n, B_n u_n\rangle = \beta_n + \nu_n\alpha_n = 0$$

eliminates both $L_n\ddot{u}_n$ and the normalization term $c\,u_n$. Solving the remaining scalar equation for ξ_n gives the result. $\square$

Corollary 16.4 (Refined filtered confluence)**.** *Because the Einstein QNM branch stays fixed while the massive branch moves, their gap satisfies*

$$\boxed{\delta_m = \nu_n m + \frac{1}{2}\xi_n m^2 + O(m^3), \qquad \frac{1}{\delta_m} = \frac{1}{\nu_n m} - \frac{\xi_n}{2\nu_n^2} + O(m).}$$

The corresponding divided exponential has the expansion

$$\boxed{\frac{e^{i\omega_n(m)t} - e^{i\omega_n t}}{m} = e^{i\omega_n t}\left[i\nu_n t + m\left(\frac{i\xi_n t}{2} - \frac{\nu_n^2 t^2}{2}\right) + O(m^2)\right].}$$

Remark 16.5 (Scalar mass specialization). For the bulk scalar operator

$$L_m = D^2 + \omega^2 - V_2 - mf,$$

one has $A = -f$, $A' = 0$, $L' = 2\omega$, and $L'' = 2$. Accordingly, the bulk part of Theorem 16.3 reduces to

$$\xi_n = \frac{2\langle \widetilde{u}_n, B_n H_n B_n u_n\rangle - 2\nu_n^2\langle \widetilde{u}_n, u_n\rangle}{\alpha_n}.$$

For a physical QNM theorem these expressions remain augmented by the derivatives of the transparent endpoint maps. No numerical value of ξ_n is asserted here.

The tangent vector $\dot{u}_n$ depends on normalization, but its class

$$\boxed{[\dot{u}_n] \in D/\mathbb{C}u_n}$$

is canonical and obeys

$$L_n\dot{u}_n = -(A_n + \nu_n L_n')u_n.$$

In the axial Bach filtration its carrier projection is $-\alpha_n/\beta_n = -1/\kappa_n$. Thus the generalized QNM is the normalization-independent tangent class of the massive QNM branch. Proposition 16.2 also shows that it can be reconstructed from the scalar QNM, its adjoint, and the finite part of the ordinary Regge–Wheeler resolvent; the six-state integration remains an independent certificate rather than an analytic necessity.

Proposition 16.6 (Augmented QNM Hellmann–Feynman formula)**.** *For a scalar realization*

$$L(\omega, m)y = [\partial_x^2 + Q(x,\omega,m)]y = 0,$$

let y_- and y_+ be the left and right Jost solutions and choose the Evans orientation $a = W(y_-, y_+)$. For any parameter p,

$$\boxed{\partial_p a = \int_{x_-}^{x_+} y_-(x)\,\partial_p Q(x,\omega,m)\,y_+(x)\,dx + B_p,}$$

where B_p is the transparent-boundary contribution (equivalently, the endpoint-normalization term) in the augmented pairing. At a simple QNM,

$$\boxed{\nu_n = -\frac{\partial_m a}{\partial_\omega a} = -\frac{\langle \widetilde{u}_n, A_n u_n \rangle_{\mathrm{aug}}}{\langle \widetilde{u}_n, L_n' u_n \rangle_{\mathrm{aug}}}.}$$

For $Q = \omega^2 - V_2 - mf$, this is

$$\boxed{\nu_n = -\frac{-\displaystyle\int f\, u_n^2 \, dx + B_m}{2\omega_n \displaystyle\int u_n^2 \, dx + B_\omega}.}$$

The integrals are bilinear QNM integrals, not positive L^2 norms.

Proof. Differentiate the two scalar equations, apply the Lagrange identity, and integrate between the endpoint cuts. The bulk term is the displayed integral; differentiating the analytic transparent maps supplies B_p. At a QNM the two Jost solutions are proportional, and the proportionality cancels from $-a_m/a_\omega$. The augmented formula is the same identity written invariantly. □

16.1 Spectral acceleration and the second critical jet

The shear quotient of Theorem 15.14 is a convenient representative of a canonical meromorphic one-form.

Theorem 16.7 (First and second spectral-flow forms)**.** *On an analytic massive Evans domain, define*

$$\Theta_1(\omega) = -\,\partial_m \log a(\omega, m)\big|_{m=0}\, d\omega, \qquad \Theta_2(\omega) = -\,\partial_m^2 \log a(\omega, m)\big|_{m=0}\, d\omega.$$

At a simple QNM branch

$$\omega_n(m) = \omega_n + \nu_n m + \frac{1}{2}\xi_n m^2 + O(m^3),$$

their singular parts are

$$\boxed{\Theta_1 = \left[\frac{\nu_n}{\omega - \omega_n} + \text{holomorphic}\right] d\omega,}$$

and

$$\boxed{\Theta_2 = \left[\frac{\nu_n^2}{(\omega - \omega_n)^2} + \frac{\xi_n}{\omega - \omega_n} + \text{holomorphic}\right] d\omega.}$$

In particular,

$$\operatorname*{Res}_{\omega=\omega_n} \Theta_1 = \nu_n, \qquad \operatorname*{Res}_{\omega=\omega_n} \Theta_2 = \xi_n.$$

Multiplication $a \mapsto v(\omega, m)a$ by an analytic nonzero Evans unit changes each Θ_j only by a holomorphic one-form. Away from $\omega = 0$,

$$\boxed{-\frac{2}{i\omega}\frac{b_{\mathrm{B}}}{a}\, d\omega = \Theta_1 + \text{holomorphic one-form.}}$$

Proof. Factor the simple divisor as

$$a(\omega,m) = u(\omega,m)\big(\omega-\omega_n(m)\big), \qquad u(\omega_n,0)\neq 0.$$

One and two mass derivatives of the logarithm of the second factor give respectively

$$-\partial_m \log(\omega-\omega_n(m))\big|_0 = \frac{\nu_n}{\omega-\omega_n},$$

and

$$-\partial_m^2 \log(\omega-\omega_n(m))\big|_0 = \frac{\nu_n^2}{(\omega-\omega_n)^2} + \frac{\xi_n}{\omega-\omega_n}.$$

The factor u, and any further Evans unit v, contributes only holomorphic one-forms. The final identity follows by multiplying Theorem 15.14 by $-2/(i\omega)$. □

Corollary 16.8 (Weighted velocity and acceleration moments)**.** *Let φ be analytic on and inside a positively oriented contour Γ containing only simple QNMs and no boundary zero. Then*

$$\boxed{\frac{1}{2\pi i}\oint_\Gamma \varphi(\omega)\Theta_1(\omega) = \sum_{\omega_n\in\operatorname{int}\Gamma} \varphi(\omega_n)\nu_n,}$$

and

$$\boxed{\frac{1}{2\pi i}\oint_\Gamma \varphi(\omega)\Theta_2(\omega) = \sum_{\omega_n\in\operatorname{int}\Gamma} \left[\varphi(\omega_n)\xi_n + \varphi'(\omega_n)\nu_n^2\right].}$$

Thus the zeroth moments sum respectively the velocities and accelerations, while the first acceleration moment is

$$\frac{1}{2\pi i}\oint_\Gamma \omega\Theta_2 = \sum_n(\omega_n\xi_n+\nu_n^2).$$

Proof. Apply the residue theorem to the principal parts in Theorem 16.7. For the second-order pole, $\operatorname{Res}[\varphi(\omega)(\omega-\omega_n)^{-2}d\omega] = \varphi'(\omega_n)$. □

Theorem 16.9 (Evans acceleration and second-jet Laurent coefficients)**.** *At a simple moving QNM,*

$$\boxed{\xi_n = -\left.\frac{a_{mm}+2\nu_n a_{\omega m}+\nu_n^2 a_{\omega\omega}}{a_\omega}\right|_{(\omega_n,0)}.}$$

The quotient is invariant under multiplication of a by an analytic nonzero unit. It equals the augmented finite-part expression in Theorem 16.3.

For

$$J_2(\omega) = \frac{1}{2}\,\partial_m^2 R_m(\omega)\big|_{m=0}, \qquad R_m(\omega) = \frac{P_n(m)}{\omega-\omega_n(m)} + H_n(\omega,m),$$

the singular part is

$$\boxed{J_2(\omega) = \frac{\nu_n^2 P_n}{(\omega-\omega_n)^3} + \frac{\nu_n\dot P_n + \frac{1}{2}\xi_n P_n}{(\omega-\omega_n)^2} + \frac{\frac{1}{2}\ddot P_n}{\omega-\omega_n} + \text{holomorphic.}}$$

If $\nu_n = 0$ but $\xi_n \neq 0$, the first jet has no double pole from resonance motion, whereas J_2 has the double coefficient $\xi_n P_n/2$.

Proof. Differentiate $a(\omega_n(m), m) = 0$ twice:

$$a_\omega \xi_n + a_{\omega\omega}\nu_n^2 + 2a_{\omega m}\nu_n + a_{mm} = 0.$$

Under $a \mapsto va$, every extra term in the numerator contains either a or $a_\omega \nu_n + a_m$; both vanish on the moving zero. The remaining numerator and a_ω acquire the same factor v. The operator expression follows from the twice-differentiated QNM equation, as proved in Theorem 16.3.

For the Laurent formula, expand

$$P_n(m) = P_n + m\dot{P}_n + \frac{1}{2}m^2\ddot{P}_n + O(m^3)$$

and

$$\frac{1}{\omega - \omega_n(m)} = \frac{1}{z} + \frac{\nu_n m + \frac{1}{2}\xi_n m^2}{z^2} + \frac{\nu_n^2 m^2}{z^3} + O(m^3), \qquad z = \omega - \omega_n.$$

The coefficient of m^2 is precisely J_2. ☐

Corollary 16.10 (Second-order isolated resonance contribution)**.** *The isolated QNM contour contribution to J_2 is*

$$\boxed{e^{i\omega_n t}\left[\frac{1}{2}\ddot{P}_n + it\,\nu_n\dot{P}_n + \left(\frac{it\,\xi_n}{2} - \frac{t^2\nu_n^2}{2}\right)P_n\right].}$$

The $t^2 e^{i\omega_n t}$ coefficient records ν_n^2, while the $te^{i\omega_n t}$ coefficient mixes acceleration with first projector motion.

Proof. Take one half of the second mass derivative of $e^{i\omega_n(m)t}P_n(m)$ at $m = 0$. ☐

Corollary 16.11 (Damped Jordan envelope)**.** *With the convention of this paper, a damped QNM has $\omega_n = \omega_R + i\gamma$, $\gamma > 0$. Hence the first-jet polynomial has envelope*

$$t|e^{i\omega_n t}| = te^{-\gamma t},$$

and

$$\boxed{t_{\max} = \frac{1}{\gamma}, \qquad \max_{t\geq 0} te^{-\gamma t} = \frac{1}{e\gamma}.}$$

For the certified mode, $\gamma = 0.0889623156889357$ in the $M = 1$ normalization, giving $t_{\max} \simeq 11.241M$ and maximal envelope factor $4.135M$. Thus the isolated Jordan contribution is transiently enhanced but decays; this is not a global retarded stability theorem.

Corollary 16.12 (Second-order reflection audit)**.** *The reflected branch satisfies*

$$\nu_n^\sharp = -\overline{\nu_n}, \qquad \xi_n^\sharp = -\overline{\xi_n}.$$

Consequently a reflected pair has purely imaginary acceleration sum, whereas

$$(\omega_n\xi_n + \nu_n^2) + (\omega_n^\sharp\xi_n^\sharp + (\nu_n^\sharp)^2)$$

is real. These are exact audits for future symmetric-contour moment calculations; no numerical acceleration contour is claimed here.

On a fixed gauge-reduced domain, let

$$\mathcal{E}_m(\omega) = \mathcal{E}(\omega) + m\mathcal{A}.$$

The metric–metric block of the inverse parent Hessian is

$$G^{\mathrm{W}}_{hh}(\omega) = \frac{1}{4\alpha_{\mathrm{W}}}\mathcal{E}(\omega)^{-1}\mathcal{A}\mathcal{E}(\omega)^{-1}.$$

Theorem 16.13 (Critical mass derivative of the metric Green operator)**.** *Where the inverses exist,*

$$\boxed{G^{\mathrm{W}}_{hh}(\omega) = -\frac{1}{4\alpha_{\mathrm{W}}}\ \partial_m\mathcal{E}_m(\omega)^{-1}\big|_{m=0}\,.}$$

No commutativity between $\mathcal{E}$ and $\mathcal{A}$ is required. If, on a fixed-domain meromorphic realization,

$$\mathcal{E}_m(\omega)^{-1} = \frac{P_n(m)}{\omega - \omega_n(m)} + H_n(\omega, m),$$

then

$$\boxed{G_{-2} = -\frac{\nu_n}{4\alpha_{\mathrm{W}}}P_n, \qquad G_{-1} = -\frac{1}{4\alpha_{\mathrm{W}}}\dot{P}_n.}$$

Proof. Differentiate $\mathcal{E}_m\mathcal{E}_m^{-1} = I$:

$$\partial_m\mathcal{E}_m^{-1} = -\mathcal{E}_m^{-1}\mathcal{A}\mathcal{E}_m^{-1}.$$

Differentiating the simple-pole expansion gives

$$\partial_m\mathcal{E}_m^{-1}\big|_0 = \frac{\nu_n P_n}{(\omega-\omega_n)^2} + \frac{\dot{P}_n}{\omega-\omega_n} + \dot{H}_n(\omega, 0).$$

□

The corresponding exact finite-mass secant identity is

$$\boxed{\frac{\mathcal{E}^{-1} - \mathcal{E}_m^{-1}}{m} = \mathcal{E}^{-1}\mathcal{A}\mathcal{E}_m^{-1} \longrightarrow \mathcal{E}^{-1}\mathcal{A}\mathcal{E}^{-1}.}$$

It follows from the noncommutative resolvent identity and makes the Weyl metric propagator the confluent difference quotient of the massless and massive second-order propagators.

For

$$P_n = \frac{u_n \otimes \widetilde{u}_n}{\alpha_n}, \qquad \alpha_n = \langle \widetilde{u}_n, \partial_\omega\mathcal{E}(\omega_n)u_n\rangle, \qquad \beta_n = \langle \widetilde{u}_n, \mathcal{A}u_n\rangle,$$

the differentiated eigenvalue equation gives

$$\alpha_n\nu_n + \beta_n = 0.$$

Hence

$$G_{-2} = \frac{1}{4\alpha_{\mathrm{W}}}\frac{\beta_n}{\alpha_n^2}u_n \otimes \widetilde{u}_n = -\frac{i}{2\alpha_{\mathrm{W}}\omega_n}\kappa_n P_n.$$

Thus parent block inversion, mass differentiation, the resonant overlap, and the projective selector give the same nonzero double coefficient. This metric–metric block should not be confused with the full triangular root-space coefficient in Theorem 17.10: the latter is nilpotent under composition on the two-layer extension space. The projected rank-one coefficient C_{-2} or G_{-2} need not itself square to zero on the projected source/response space.

Theorem 16.14 (Isolated parent-resonance contribution)**.** *For a small contour enclosing only the massive QNM branch,*

$$\boxed{\mathcal{U}_n^{\mathrm{W}}(t) = -\frac{e^{i\omega_n t}}{4\alpha_{\mathrm{W}}}\left[\dot{P}_n + it\,\nu_n P_n\right].}$$

For a compact source F and an observation functional Λ, the coefficient of $te^{i\omega_n t}$ is

$$\boxed{A_t(F,\Lambda) = -\frac{i\nu_n}{4\alpha_{\mathrm{W}}\alpha_n}\Lambda(u_n)\langle\widetilde{u}_n, F\rangle.}$$

It is present exactly when the source excites the adjoint resonant channel and the observable sees the Einstein QNM profile.

Proof. Differentiate the ordinary isolated-pole contribution $e^{i\omega_n(m)t}P_n(m)$ and apply Theorem 16.13. The second formula follows from the rank-one expression for P_n. □

The polynomial term is Einstein-shaped even though the generalized vector has a nonzero carrier quotient:

$$e^{i\omega_n t}(V_1 + itV_0).$$

The simple-pole coefficient contains the deformation of the mode shape and normalization. Explicitly,

$$\dot{P}_n = \frac{\dot{u}_n \otimes \widetilde{u}_n + u_n \otimes \dot{\widetilde{u}}_n}{\alpha_n} - \frac{\dot{\alpha}_n}{\alpha_n}P_n.$$

As before, this is an isolated local contour theorem, not a global retarded contour deformation.

16.2 Analytic observable transfer and detector normal form

Let $\mathcal{F}$ be a source space, X the reduced equation space, Y the reconstructed response space, and $\mathcal{H}$ an observable coefficient space. Consider analytic maps

$$S(\omega) : \mathcal{F} \longrightarrow X, \qquad \mathcal{O}(\omega) : Y \longrightarrow \mathcal{H}.$$

The map S inserts a declared source into the reduced equations, while $\mathcal{O}$ may include metric or curvature reconstruction followed by an observation functional. The following statement is algebraic in the local meromorphic germ. It does not assume that either map has already been identified with an asymptotic detector channel.

Theorem 16.15 (Observable transfer)**.** *Suppose*

$$G(\omega) = \frac{G_{-2}}{z^2} + \frac{G_{-1}}{z} + G_0(\omega), \qquad z = \omega - \omega_n,$$

and $S, \mathcal{O}$ are analytic at ω_n. For

$$\mathcal{H}(\omega) = \mathcal{O}(\omega)G(\omega)S(\omega)$$

the two singular coefficients are

$$\boxed{\mathcal{H}_{-2} = \mathcal{O}_0 G_{-2} S_0,}$$

$$\boxed{\mathcal{H}_{-1} = \mathcal{O}_0 G_{-1} S_0 + \mathcal{O}_1 G_{-2} S_0 + \mathcal{O}_0 G_{-2} S_1.}$$

For the parent pure-Weyl metric response,

$$G_{-2} = -\frac{\nu_n}{4\alpha_{\mathrm{W}}\alpha_n}u_n \otimes \widetilde{u}_n,$$

and hence

$$\boxed{\mathcal{H}_{-2} = -\frac{\nu_n}{4\alpha_{\mathrm{W}}\alpha_n}\big(\mathcal{O}_0 u_n\big) \otimes \big(\widetilde{u}_n S_0\big).} \tag{13}$$

Because $\nu_n \neq 0$ at the certified QNM, the measured channel retains the order-two pole if and only if

$$\mathcal{O}(\omega_n)u_n \neq 0, \qquad \widetilde{u}_n S(\omega_n) \neq 0.$$

Proof. Write

$$\mathcal{O} = \mathcal{O}_0 + z\mathcal{O}_1 + O(z^2), \qquad S = S_0 + zS_1 + O(z^2),$$

and multiply the three Laurent series. Only the zeroth-order endpoint values can multiply G_{-2} into the z^{-2} coefficient. The three ways to obtain z^{-1} give the displayed simple coefficient. Equation (13) follows from the rank-one parent coefficient. A rank-one tensor is nonzero exactly when both of its factors are nonzero. □

Thus analytic frequency dependence in source insertion, reconstruction, or measurement changes the accompanying simple pole but cannot cancel the leading pole except by resonant source or observation annihilation. The corollary after Theorem 13.2 certifies nonannihilation for at least one local reconstructed metric/curvature observable. It does not yet certify a particular strain or Newman–Penrose observable at $\mathscr{I}^+$, nor a specified astrophysical source.

Theorem 16.16 (Real repeated-root waveform)**.** *Write the certified member of a reflected pair as*

$$\omega_n = -\Omega + i\gamma, \qquad \gamma > 0.$$

The real form of an arbitrary isolated critical pair contribution is

$$\boxed{h(t) = e^{-\gamma t}\left[(a_0 + a_1 t)\cos(\Omega t) + (b_0 + b_1 t)\sin(\Omega t)\right],} \tag{14}$$

with real coefficients a_j, b_j. If

$$Q = \partial_t^2 + 2\gamma\partial_t + (\gamma^2 + \Omega^2),$$

then

$$Q^2 h = 0,$$

while $Qh \neq 0$ unless $a_1 = b_1 = 0$. Thus the Jordan contribution is a repeated-root damped oscillation.

Proof. A complex local contribution has the form

$$e^{(-\gamma - i\Omega)t}(A + Ct).$$

Adding its reflected conjugate and resolving the real and imaginary parts of A and C gives (14). The characteristic polynomial of Q is

$$(\lambda + \gamma)^2 + \Omega^2,$$

so its two roots are $-\gamma \pm i\Omega$. Multiplication by t creates the length-two generalized solutions. Hence Q^2 annihilates the four-dimensional real span, while Q removes only the ordinary part and leaves an ordinary damped sinusoid when the generalized coefficient is nonzero. □

Theorem 16.17 (Uniform divided-difference template)**.** *Let two nearby resonances be separated by*

$$\delta = \omega_m - \omega_0.$$

Define

$$\Phi_0(t;\delta) = e^{i\omega_0 t}, \qquad \Phi_1(t;\delta) = e^{i\omega_0 t}\frac{e^{i\delta t}-1}{\delta},$$

where $\Phi_1(t;0) = it\, e^{i\omega_0 t}$ *by analytic continuation. Every two-mode expression*

$$r_0 e^{i\omega_0 t} + r_1 e^{i(\omega_0+\delta)t}$$

has the exact form

$$\boxed{A\Phi_0 + B\Phi_1, \qquad A = r_0 + r_1, \quad B = \delta r_1.}$$

For a filtered confluence the physical coefficients A, B *remain finite as* $\delta \to 0$*, even though* $r_0, r_1 = O(\delta^{-1})$*. The limiting signal is*

$$e^{i\omega_0 t}\big[A(0) + iB(0)t\big].$$

Proof. Substitution of $A = r_0 + r_1$ and $B = \delta r_1$ gives the separated two-mode expression identically. The apparent singularity at $\delta = 0$ is removable because

$$\frac{e^{i\delta t}-1}{\delta} \longrightarrow it.$$

The confluent-projector theorem supplies the $O(\delta^{-1})$ separated residues and the finite combined coefficients. □

The dimensionless crossover parameter is

$$\eta = \frac{|\delta|}{\gamma} \simeq \frac{|\nu_n m|}{\gamma}.$$

When $\eta \ll 1$, the separation is below the damping width and $\Phi_1 \simeq it\, e^{i\omega_0 t}$ over the useful ringdown lifetime. When $\eta = O(1)$, the divided-difference template is the regular crossover description. When $\eta \gg 1$, the two simple modes are spectrally resolved. These are local linewidth comparisons, not a detector-sensitivity theorem.

Proposition 16.18 (Finite norm of the Jordan component)**.** *For*

$$g(t) = Ct\, e^{(-\gamma - i\Omega)t}, \qquad \gamma > 0,$$

one has

$$\boxed{\int_0^\infty |g(t)|^2\, dt = \frac{|C|^2}{4\gamma^3}, \qquad \int_0^\infty |\dot g(t)|^2\, dt = \frac{|C|^2(\gamma^2+\Omega^2)}{4\gamma^3}.}$$

Thus the isolated polynomial mode has finite integrated amplitude and finite derivative norm.

Proof. Use

$$\int_0^\infty t^k e^{-2\gamma t}\, dt = \frac{k!}{(2\gamma)^{k+1}}, \qquad k = 0, 1, 2,$$

and

$$\dot g = C\big[1 - (\gamma + i\Omega)t\big]e^{(-\gamma - i\Omega)t}.$$

The constant and linear terms in the derivative norm cancel after integration, leaving the displayed result. □

Remark 16.19 (Observational degeneracy and remaining bridge). Since

$$\partial_\omega e^{i\omega t} = it\, e^{i\omega t},$$

the Jordan quadrature is tangent to changes in oscillation frequency and damping. On a finite noisy interval it can also be covariant with two nearby ordinary modes or a change in ringdown start time. The uniform template provides a well-conditioned parameterization, but no detectability or parameter-estimation claim is made here.

A true asymptotic gravitational-wave statement requires analytic maps

$$S_{\mathrm{phys}}(\omega), \qquad \mathcal{O}_{\mathscr{I}^+}(\omega)$$

and separate proofs that

$$\widetilde{u}_n S_{\mathrm{phys}}(\omega_n) \neq 0, \qquad \mathcal{O}_{\mathscr{I}^+}(\omega_n) u_n \neq 0.$$

It also requires the global causal contour control excluded in Section 14. Theorem 16.15 isolates these as the only source/observation annihilation gates for the principal local pole.

16.3 Finite-time coherent forcing of the isolated resonance

The previous results determine a causal local kernel after source and observation projection. Write its critical and ordinary parts as

$$g_{\mathrm{W}}(t) = \Theta(t) C\, t\, e^{-\gamma t} e^{-i\Omega t}, \qquad g_{\mathrm{E}}(t) = \Theta(t) A\, e^{-\gamma t} e^{-i\Omega t}.$$

The constants C and A include the relevant residues and declared source/observation overlaps. These kernels represent only the isolated resonance contribution.

Theorem 16.20 (Switched-on coherent drive)**.** *Let*

$$F(t) = F_0 \Theta(t) e^{-i\Omega_d t}, \qquad \Delta = \Omega - \Omega_d, \qquad \lambda = \gamma + i\Delta.$$

The causal convolutions with the critical and ordinary kernels are

$$\boxed{x_{\mathrm{W}}(t) = C F_0 e^{-i\Omega_d t} \frac{1 - (1 + \lambda t) e^{-\lambda t}}{\lambda^2},}$$

$$\boxed{x_{\mathrm{E}}(t) = A F_0 e^{-i\Omega_d t} \frac{1 - e^{-\lambda t}}{\lambda}.}$$

At real-frequency tuning, $\Omega_d = \Omega$*,*

$$x_{\mathrm{W}}(t) = \frac{1}{2} C F_0 t^2 e^{-i\Omega t} + O(t^3), \qquad x_{\mathrm{E}}(t) = A F_0 t e^{-i\Omega t} + O(t^2).$$

For $\gamma > 0$*, both responses saturate:*

$$x_{\mathrm{W}}(t) \longrightarrow \frac{C F_0}{(\gamma + i\Delta)^2} e^{-i\Omega_d t}, \qquad x_{\mathrm{E}}(t) \longrightarrow \frac{A F_0}{\gamma + i\Delta} e^{-i\Omega_d t}.$$

Proof. With $\tau = t - s$,

$$x_{\mathrm{W}}(t) = CF_0 e^{-i\Omega_d t}\int_0^t \tau e^{-\lambda\tau}\,d\tau,$$

and

$$x_{\mathrm{E}}(t) = AF_0 e^{-i\Omega_d t}\int_0^t e^{-\lambda\tau}\,d\tau.$$

The two elementary integrals give the displayed formulas. Taylor expansion at $t = 0$ gives the quadratic and linear buildup laws, and $\Re\lambda = \gamma > 0$ gives the long-time limits. $\square$

The critical steady-state amplitude is

$$|H_{\mathrm{W}}(\Omega_d)| = \frac{|C|}{\gamma^2 + \Delta^2},$$

whereas the ordinary amplitude is

$$|H_{\mathrm{E}}(\Omega_d)| = \frac{|A|}{\sqrt{\gamma^2 + \Delta^2}}.$$

Absolute amplitudes cannot be compared without fixing the dimensions and normalizations of A and C, but the critical damping dependence has one additional inverse power. The half-power detuning of the critical channel is

$$\boxed{|\Delta|_{\mathrm{hp}} = \gamma\sqrt{\sqrt{2} - 1} \simeq 0.644\,\gamma,}$$

instead of $|\Delta|_{\mathrm{hp}} = \gamma$ for the ordinary Lorentzian.

Proposition 16.21 (Phase-matched impulse train)**.** *Let N equal impulses be separated by*

$$T = \frac{2\pi}{\Omega}, \qquad q = e^{-\gamma T},$$

and observe immediately one period after the last pulse, so their ages are $T, 2T, \ldots, NT$. The phase-aligned critical coefficient is proportional to

$$T\sum_{j=1}^{N} jq^j = \boxed{T\frac{q[1 - (N+1)q^N + Nq^{N+1}]}{(1-q)^2}.}$$

If $NT \ll 1/\gamma$, this becomes

$$T\frac{N(N+1)}{2} + O(\gamma T^2 N^3),$$

whereas the corresponding ordinary dimensionless sum is $N + O(\gamma T N^2)$. Thus the coherent undamped window has critical N^2 buildup and ordinary N buildup, with their separate single-pulse normalizations understood.

Proof. Phase matching gives $e^{-i\Omega jT} = 1$. The age-jT critical kernel contributes $CjTq^j$. Differentiating the finite geometric sum $\sum_{j=0}^{N} q^j$ gives the closed form. Expanding q^j uniformly when $\gamma NT \ll 1$ gives the stated asymptotics. $\square$

For the certified mode,

$$\Omega \simeq 0.3736716844\, M^{-1}, \qquad \gamma \simeq 0.0889623157\, M^{-1}.$$

Consequently,

$$\tau_{\text{damp}} = \gamma^{-1} \simeq 11.24M, \qquad T_{\text{osc}} = \frac{2\pi}{\Omega} \simeq 16.81M,$$

$$e^{-\gamma T_{\text{osc}}} \simeq 0.224, \qquad Q_{\text{mode}} = \frac{\Omega}{2\gamma} \simeq 2.10.$$

The certified mode is therefore strongly damped: a long pulse train loses most of each contribution before the next full cycle. In the local model, useful matched forcing is concentrated within one or a few damping times, not over a large number of cycles.

Theorem 16.22 (Optimal finite-window coefficient-space drive)**.** *Fix a target time $T > 0$ and the declared source-coefficient budget*

$$\int_0^T |F(s)|^2\, ds \le E.$$

For

$$x(T) = \int_0^T g_{\mathrm{W}}(T-s)F(s)\, ds,$$

one has

$$\boxed{|x(T)| \le \frac{|C|\sqrt{E}}{2\gamma^{3/2}} \sqrt{1 - e^{-2\gamma T}(1 + 2\gamma T + 2\gamma^2 T^2)}.}$$

Equality is attained by

$$\boxed{F_{\text{opt}}(s) = c_T\, \overline{g_{\mathrm{W}}(T-s)} \propto (T-s)e^{-\gamma(T-s)}e^{+i\Omega(T-s)},}$$

with c_T chosen to saturate the budget. As $T \to \infty$, the bound tends to

$$\frac{|C|\sqrt{E}}{2\gamma^{3/2}}.$$

Proof. Cauchy–Schwarz gives

$$|x(T)| \le \sqrt{E} \left[\int_0^T |g_{\mathrm{W}}(T-s)|^2\, ds \right]^{1/2}.$$

After $\tau = T - s$, the remaining integral is

$$|C|^2 \int_0^T \tau^2 e^{-2\gamma\tau}\, d\tau = \frac{|C|^2}{4\gamma^3} \left[1 - e^{-2\gamma T}(1 + 2\gamma T + 2\gamma^2 T^2)\right].$$

Equality in Cauchy–Schwarz holds precisely for the displayed conjugate matched kernel, up to an overall phase and normalization. □

Remark 16.23 (Physical and nonlinear boundary). The optimizer is exact in the declared one-dimensional coefficient-space L^2 norm. A spacetime realization would additionally have to match the adjoint radial profile, axial parity, and $\ell = 2$ angular pattern, and would require a physical stress-energy or incident-wave source map. The budget E has not been identified with invariant gravitational energy. Precision improves coupling efficiency but does not remove the physical source scale.

All formulas in this subsection are linear. Frequency shifts, backreaction, additional modes, and nonlinear mode coupling lie outside the result. Most importantly, the isolated kernel has not yet been promoted to the complete global retarded Schwarzschild response.

Theorem 16.24 (Spectral contact controls critical pole order)**.** *Let*

$$R_m(\omega) = \frac{P_n(m)}{\omega - \omega_n(m)} + H_n(\omega, m)$$

be an analytic family of simple-pole resolvents and suppose the first nonzero mass derivative of the resonance occurs at order q:

$$\omega_n(m) = \omega_n + \frac{\nu_{n,q}}{q!} m^q + O(m^{q+1}), \qquad \nu_{n,q} \neq 0.$$

For $p \geq 0$, define the normalized critical jet

$$J_p(\omega) = \frac{(-1)^p}{p!} \, \partial_m^p R_m(\omega)\big|_{m=0}.$$

Then

$$\boxed{\operatorname{poleord}_{\omega_n} J_p \leq \left\lfloor \frac{p}{q} \right\rfloor + 1.}$$

For $p < q$, resonance motion produces no pole enhancement, although derivatives of $P_n(m)$ may retain a simple pole. The first enhanced pole appears at $p = q$, is second order, and has coefficient

$$\boxed{\operatorname{Coeff}_{(\omega-\omega_n)^{-2}} J_q = \frac{(-1)^q}{q!} \nu_{n,q} P_n.}$$

More generally, for $p = kq$ the highest possible pole has order $k+1$ and coefficient

$$\boxed{\operatorname{Coeff}_{(\omega-\omega_n)^{-(k+1)}} J_{kq} = (-1)^{kq} \left(\frac{\nu_{n,q}}{q!} \right)^k P_n.}$$

Proof. Put $z = \omega - \omega_n$ and $\delta(m) = \omega_n(m) - \omega_n$. The moving denominator has the formal expansion

$$\frac{1}{z - \delta(m)} = \sum_{k \geq 0} \frac{\delta(m)^k}{z^{k+1}}.$$

Because $\operatorname{ord}_{m=0} \delta = q$, a term of pole order $k+1$ can enter the p-th derivative only if $kq \leq p$. This proves the bound. At $p = q$, the coefficient of $m^q z^{-2}$ is $(\nu_{n,q}/q!) P_n$; at $p = kq$, the coefficient of $m^{kq} z^{-(k+1)}$ is $(\nu_{n,q}/q!)^k P_n$. Multiplication by the normalization $(-1)^p$ of J_p gives the two displayed coefficients. □

Corollary 16.25 (Transverse higher jets)**.** *For the linear pencil $L + mA$, repeated differentiation gives*

$$R(AR)^p = \frac{(-1)^p}{p!} \, \partial_m^p (L + mA)^{-1}\big|_{m=0}.$$

If $\beta_n \neq 0$, then $q = 1$, $\nu_n = -\beta_n/\alpha_n$, and the pole order is $p + 1$, with leading coefficient

$$\frac{\beta_n^p}{\alpha_n^{p+1}} u_n \otimes \widetilde{u}_n.$$

Its leading isolated-contour polynomial is

$$\frac{(it)^p}{p!} e^{i\omega_n t} \frac{\beta_n^p}{\alpha_n^{p+1}} u_n \otimes \widetilde{u}_n.$$

Pure Weyl gravity is the transverse first-jet case $p = q = 1$.

The contact order q classifies the critical response. Transverse criticality ($q = 1$) is visible in the first jet. Tangential criticality ($q = 2$) has no first-jet double pole but acquires one in the second jet, not automatically a triple pole. Higher contact $q \geq 3$ is invisible to pole enhancement in the first $q - 1$ mass derivatives.

For further Schwarzschild overtones, the universal criterion replaces a full six-state Smith calculation by two validated scalar tasks: certify a simple Regge–Wheeler QNM and enclose the augmented overlap $\beta_n = \langle \widetilde{u}_n, A_n u_n \rangle$ away from zero. The Hellmann–Feynman identity

$$\omega_n'(0) = -\frac{\langle \widetilde{u}_n, A_n u_n \rangle_{\text{aug}}}{\langle \widetilde{u}_n, L_n' u_n \rangle_{\text{aug}}}$$

then supplies the mass velocity, pole order, and generalized root length. Proposition 16.6 reduces the overlap to a bilinear scalar Wronskian integral plus explicit endpoint terms. A validated evaluation for several overtones would decide whether the certified reflected EP2 pair is the first member of a defective tower; no such tower is claimed here. Complementarily, Theorem 15.14 permits a validated contour integral of b_{B}/a to sum many selectors at once. A nonzero sum proves that at least one enclosed QNM is defective, but a zero sum is inconclusive because nonzero selectors may cancel. Nested contours can therefore guide, but do not replace, the mode-by-mode augmented-overlap certificates.

17 Two-parameter filtration-protected unfolding

Put

$$\nu_n = \left. \frac{d\omega_n}{dm} \right|_{m=0} = \frac{2i}{\omega_n} \kappa_n \neq 0.$$

The massless divisor remains at ω_n, while the massive divisor is

$$\omega_m = \omega_n + \nu_n m + O(m^2).$$

Proposition 17.1 (Generic versus filtration-preserving splitting)**.** *For a general first-order perturbation of*

$$T_0(z) = \begin{pmatrix} z & -1 \\ 0 & z \end{pmatrix},$$

a nonzero lower-left entry produces Newton–Puiseux splitting $\Delta = O(m^{1/2})$. If the perturbation preserves the Einstein submodule, that entry vanishes identically and the two roots are analytic. The physical Einstein–Weyl mass deformation is of the latter type: the Einstein root remains fixed and the carrier root moves with nonzero velocity ν_n.

Theorem 17.2 (Invariant two-parameter unfolding)**.** *Let $F(\omega, m, \epsilon)$ be the determinant of the reduced two-spin-two connection block, or the complete determinant divided by the local spin-one unit, and put $z = \omega - \omega_n$. At the critical point,*

$$F = F_\omega = 0, \qquad F_{\omega\omega} \neq 0.$$

The exact Einstein branch gives $F(\omega_n, m, 0) = 0$, hence $F_m = 0$ at the origin. Define

$$\boxed{\nu_n = -\frac{2F_{\omega m}}{F_{\omega\omega}}, \qquad c_n = \frac{2F_\epsilon}{F_{\omega\omega}}.}$$

For a declared filtration-breaking direction with $F_\epsilon \neq 0$, the Weierstrass-reduced characteristic equation is

$$\boxed{z^2 - \nu_n m z + c_n \epsilon + O(z^3, z^2 m, z m^2, z\epsilon, m\epsilon, m^3, \epsilon^2) = 0.}$$

Consequently,

$$\boxed{z_\pm = \frac{\nu_n m}{2} \pm \frac{1}{2}\sqrt{\nu_n^2 m^2 - 4c_n \epsilon} + \cdots, \qquad \Delta^2 = \nu_n^2 m^2 - 4c_n\epsilon + \cdots.}$$

The pair (ν_n, c_n) is invariant under multiplication $F \mapsto uF$ by an analytic nonzero unit.

Proof. Taylor expansion gives

$$F = \frac{1}{2}F_{\omega\omega} z^2 + F_{\omega m} z m + F_\epsilon \epsilon + \text{higher terms}.$$

Division by $F_{\omega\omega}/2$ gives the displayed characteristic equation. If $\widetilde{F} = uF$, the identities $F = F_\omega = F_m = 0$ imply

$$\widetilde{F}_{\omega\omega} = uF_{\omega\omega}, \qquad \widetilde{F}_{\omega m} = uF_{\omega m}, \qquad \widetilde{F}_\epsilon = uF_\epsilon$$

at the critical point, so both ratios are unchanged. □

Theorem 17.3 (Lidskii reverse-coupling coefficient)**.** *Let the critical full pencil have chains*

$$\mathbb{L}_n V_0 = 0, \qquad \mathbb{L}_n V_1 + \mathbb{L}_n' V_0 = 0, \qquad \mathbb{L}_n^* W_0 = 0,$$

and set

$$d_n = \left\langle W_0, \mathbb{L}_n' V_1 + \frac{1}{2}\mathbb{L}_n'' V_0 \right\rangle.$$

For a genuine length-two root $d_n \neq 0$. If $\mathbb{L} \mapsto \mathbb{L} + \epsilon B$, then

$$\boxed{c_n = \frac{\langle W_0, BV_0\rangle}{d_n}.}$$

This ratio is unchanged by a common scaling of the root chain or by $V_1 \mapsto V_1 + \lambda V_0$.

Proof. Use $z = O(\epsilon^{1/2})$ and $V = V_0 + zV_1 + z^2 V_2 + \cdots$. At order $z^2 \sim \epsilon$, pairing the expanded equation with W_0 gives

$$d_n z^2 + \epsilon\langle W_0, BV_0\rangle = 0.$$

Thus $z^2 = -c_n\epsilon$, agreeing with Theorem 17.2 at $m = 0$. The shift of V_1 changes d_n by $\lambda\langle W_0, \mathbb{L}_n' V_0\rangle = 0$. □

The forward overlap

$$\beta_n = \langle \widetilde{u}_n, K_n u_n \rangle \neq 0$$

creates the non-split carrier-to-Einstein chain. The reverse functional

$$\gamma_n(B) = \langle W_0, BV_0 \rangle$$

measures Einstein-to-carrier mixing. The physical mass deformation preserves the filtration and has $\gamma_n(B_{\text{mass}}) = 0$; a declared generic mixing direction has $\gamma_n(B) \neq 0$.

Corollary 17.4 (Exceptional parabola and branch monodromy)**.** *To leading order, the defective locus is*

$$\boxed{\epsilon_{\text{EP}}(m) = \frac{\nu_n^2}{4c_n} m^2 = \frac{F_{\omega m}^2}{2F_{\omega\omega}F_\epsilon} m^2.}$$

This is an analytic parabola in the complexified deformation space. A chosen real two-parameter slice contains a real exceptional curve only when the coefficient is compatible with its reality structure. In deformation space the parabola is tangent at the origin to the physical mass axis $\epsilon = 0$. A generic path $\epsilon = \lambda m + O(m^2)$, $\lambda \neq 0$, instead has $\Delta = O(m^{1/2})$. Encircling the discriminant changes the sign of the square root and exchanges z_+ and z_-. No such exchange occurs on the physical mass axis, where the exact filtration labels the Einstein and carrier branches analytically.

Remark 17.5 (Two meanings of transverse)*.* The massless and massive resonance divisors intersect transversely in the (ω, m)-plane because their QNM velocities differ by $\nu_n \neq 0$. By contrast, the physical mass direction $\epsilon = 0$ is tangent to the exceptional discriminant curve in the enlarged (m, ϵ)-deformation space.

Proposition 17.6 (Filtration-error threshold)**.** *If an intended mass deformation m contains a lower-left numerical or gauge contamination ϵ_{err}, recovery of the physical linear coefficient requires*

$$\boxed{|c_n \epsilon_{\text{err}}| \ll |\nu_n^2 m^2|, \qquad |\epsilon_{\text{err}}| \ll \frac{|\nu_n|^2}{|c_n|} |m|^2.}$$

If $\epsilon_{\text{err}} = O(m^p)$, then $p < 2$ is mixing dominated, $p = 2$ changes the linear coefficient, and $p > 2$ recovers ν_n asymptotically. The natural dimensionless crossover variable is

$$\boxed{\chi = \frac{4c_n \epsilon}{\nu_n^2 m^2}, \qquad \Delta = \nu_n m \sqrt{1 - \chi}}$$

in a chosen analytic square-root branch.

Proposition 17.7 (Lower-left mutation certificate)**.** *For the exact leading mutation*

$$T_{\text{mut}}(z; m, \epsilon) = \begin{pmatrix} z & -1 \\ c_n \epsilon & z - \nu_n m \end{pmatrix},$$

the following three limits distinguish filtered from generic splitting:

$$\boxed{\left. \frac{\Delta}{m} \right|_{\epsilon=0} = \nu_n, \qquad \left. \frac{\Delta^2}{\epsilon} \right|_{m=0} = -4c_n, \qquad \epsilon_{\text{EP}}(m) = \frac{\nu_n^2}{4c_n} m^2.}$$

For an analytic full pencil the equalities acquire respectively $O(m)$, $O(\epsilon)$, and $O(m^3)$ remainders. These are algebraic mutation controls; no numerical value of c_n is claimed until a concrete filtration-breaking perturbation is specified and certified.

Theorem 17.8 (Filtered critical-mass unfolding normal form)**.** *Near $(\omega_n, 0)$, the two-channel connection pencil is analytically left–right equivalent to*

$$\boxed{\mathcal{T}(z,\mu) = \begin{pmatrix} z & -1 \\ 0 & z-\mu \end{pmatrix},}$$

where $z = z(\omega, m)$ and $\mu = \mu(m)$ are analytic local coordinates,

$$z(\omega_n, 0) = 0, \qquad \partial_\omega z(\omega_n, 0) \neq 0,$$

and

$$\mu(m) = \partial_\omega z(\omega_n, 0)\, \nu_n m + O(m^2).$$

Thus the pure-Weyl EP2 is a filtration-preserving confluence whose two QNM divisors are transverse in the (ω, m)-plane. This is distinct from transversality to the exceptional discriminant in deformation space, and it is not a generic miniversal perturbation of a Jordan block.

Proof. In a finite-mass branch frame the two diagonal divisors are scalar Evans functions $a_0(\omega)$ and $a_m(\omega, m)$. Both have a simple zero at $(\omega_n, 0)$, while Theorem 15.11 says that their zero loci have distinct nonzero tangents in the mass direction. Take z to be an analytic unit multiple of a_0, and take μ to be the normalized difference of the two divisors. The off-diagonal entry in the confluent frame is a unit because the certified Smith type is $(0, 0, 2)$. Analytic invertible row and column operations normalize that unit to -1 and remove the remaining divisible entries. ☐

Theorem 17.9 (Gap-controlled projectors, nilpotent, and metrics)**.** *For the leading two-parameter normal form, choose right and left vectors*

$$v_\pm = \begin{pmatrix} 1 \\ z_\pm \end{pmatrix}, \qquad w_\pm^\mathsf{T} = \begin{pmatrix} z_\pm - \nu_n m & 1 \end{pmatrix}.$$

Then

$$w_\pm^\mathsf{T} v_\pm = 2z_\pm - \nu_n m = \pm\Delta$$

and the spectral projectors are

$$\boxed{P_\pm = \frac{v_\pm w_\pm^\mathsf{T}}{\pm\Delta}, \qquad \|P_\pm\| \asymp |\Delta|^{-1}.}$$

In the normalized critical basis,

$$\boxed{N = \lim_{(m,\epsilon)\to(0,0)} \frac{\Delta}{2}(P_+ - P_-) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.}$$

This limit is independent of the approach path through points with $\Delta \neq 0$, provided no additional spectral degeneracy is introduced.

Let

$$S = \begin{pmatrix} 1 & 1 \\ z_+ & z_- \end{pmatrix}.$$

Since $\det S = -\Delta$, *one has*

$$\operatorname{cond} S \asymp |\Delta|^{-1}.$$

For any uniformly bounded and boundedly invertible positive diagonal branch metric H_{br}, its pullback

$$H = (S^{-1})^\dagger H_{\mathrm{br}} S^{-1}$$

satisfies

$$\boxed{\operatorname{cond} H \asymp |\Delta|^{-2}.}$$

Thus the physical mass axis gives projector growth $O(|m|^{-1})$ and metric conditioning $O(|m|^{-2})$, whereas the transverse mixing axis gives $O(|\epsilon|^{-1/2})$ and $O(|\epsilon|^{-1})$.

Proof. Direct multiplication gives $T(z_\pm; m, \epsilon)v_\pm = 0$ and $w_\pm^\mathsf{T} T(z_\pm; m, \epsilon) = 0$ to leading order. The biorthogonal projector formula gives $P_\pm$. Moreover,

$$\frac{\Delta}{2}(P_+ - P_-) = \frac{1}{2}\left(v_+ w_+^\mathsf{T} + v_- w_-^\mathsf{T}\right) \longrightarrow N.$$

The smallest singular value of S is comparable to $|\Delta|$, while the largest remains bounded above and below. Pullback of a uniformly positive branch metric squares this singular-value ratio. □

Theorem 17.10 (Root-space polarization and nilpotent pole)**.** *Let*

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

with

$$L_n u_n = 0, \qquad L_n^* \widetilde{u}_n = 0, \qquad \alpha_n = \langle \widetilde{u}_n, L_n' u_n \rangle \neq 0, \qquad \beta_n = \langle \widetilde{u}_n, K_n u_n \rangle \neq 0.$$

The right and left geometric root vectors are

$$\boxed{V_0 = (u_n, 0), \qquad W_0 = (0, \widetilde{u}_n).}$$

They are self-orthogonal in the full-pencil duality:

$$\langle W_0, V_0 \rangle = 0, \qquad \langle W_0, \mathbb{L}_n' V_0 \rangle = 0.$$

The first generalized right vector has carrier component

$$-\frac{\alpha_n}{\beta_n} u_n = -\frac{1}{\kappa_n} u_n.$$

A left generalized vector can be chosen with Einstein-dual component $-(\alpha_n/\beta_n)\widetilde{u}_n$; both chains cross the filtration at first generalized order. The second-order Laurent coefficient is

$$\boxed{R_{-2} = -\frac{\beta_n}{\alpha_n^2} V_0 \otimes W_0, \qquad R_{-2}^2 = 0.}$$

Its range is the Einstein resonant line and that line lies in its kernel. Consequently R_{-2} is, up to its nonzero scalar normalization, the nilpotent generator N of the dual-number commutant.

Proof. If $(x, cu_n) \in \ker \mathbb{L}_n$, then $L_n x + cK_n u_n = 0$. Pairing with $\widetilde{u}_n$ gives $c\beta_n = 0$, hence $c = 0$. The adjoint argument gives the stated left root. For a right chain $\mathbb{L}_n V_1 + \mathbb{L}_n' V_0 = 0$, write the carrier component of V_1 as γu_n. The scalar solvability condition is $\gamma\beta_n + \alpha_n = 0$. The Laurent coefficient follows from the block inverse. Its square vanishes because $(V_0 \otimes W_0)^2 = \langle W_0, V_0 \rangle V_0 \otimes W_0 = 0$. □

Corollary 17.11 (Spectral cotangent realization)**.** *On the resonant fiber,*

$$R_{-2} : (B_2/E)^* \longrightarrow E$$

is a nonzero map between one-dimensional spaces and hence an isomorphism, up to the scalar $-\beta_n/\alpha_n^2$. The double pole is the spectral realization of the endpoint cotangent-type duality $B_2/E \simeq E^*$.

Theorem 17.12 (Canonical Krein–Jordan geometry)**.** *Let a length-two critical root space have chain basis (V_0, V_1) and nilpotent generator*

$$NV_0 = 0, \qquad NV_1 = V_0, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Every nondegenerate Hermitian form G satisfying

$$N^\dagger G = GN$$

has, in this chain basis, the form

$$G = \begin{pmatrix} 0 & b \\ b & d \end{pmatrix}, \qquad b \in \mathbb{R} \setminus \{0\}, \quad d \in \mathbb{R}.$$

The permitted chain change

$$V_1 \longmapsto V_1 - \frac{d}{2b} V_0$$

followed by a nonzero real rescaling puts the pair into the normal form

$$\boxed{N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad G \sim \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.}$$

Thus the geometric root is necessarily null. In the normalized chain basis the generalized root is also null, and

$$G(V_0, V_1) \neq 0.$$

No positive-definite Hermitian form can make a nonzero nilpotent self-adjoint.

Proof. Write

$$G = \begin{pmatrix} a & b \\ \bar{b} & d \end{pmatrix}.$$

The equation $N^\dagger G = GN$ gives $a = 0$ and $b = \bar{b}$. Nondegeneracy gives $\det G = -b^2 \neq 0$. The displayed chain shift preserves $NV_1 = V_0$ and changes the lower-right entry to zero.

Alternatively, a self-adjoint operator in a positive-definite finite-dimensional space is diagonalizable. A nilpotent self-adjoint operator has only the eigenvalue zero and must therefore vanish, contrary to $N \neq 0$. □

Corollary 17.13 (Null rank-one pole and determinant invisibility)**.** *Let $V_0^\flat = G(V_0, \cdot)$. On the critical root space the full double-pole coefficient is*

$$\boxed{R_{-2} = \gamma_n V_0 \otimes V_0^\flat, \qquad \gamma_n \neq 0, \qquad R_{-2}^2 = 0.}$$

Equivalently, the left geometric root covector satisfies

$$W_0 \propto V_0^\flat.$$

Moreover,

$$\operatorname{tr} R_{-2} = 0, \qquad \det(I + sR_{-2}) = 1, \qquad e^{sR_{-2}} = I + sR_{-2}.$$

Thus the nonzero enhanced response is invisible to ordinary eigenvalue, trace, and determinant invariants.

Proof. Both W_0 from Theorem 17.10 and $V_0^\flat$ are nonzero covectors whose kernel is the Einstein line $\mathbb{C}V_0$, so they are proportional. Nullness gives

$$(V_0 \otimes V_0^\flat)^2 = V_0^\flat(V_0)V_0 \otimes V_0^\flat = 0.$$

The remaining identities follow from square-zero nilpotency. □

Theorem 17.14 (Opposite-sign confluence criterion)**.** *Let the two separated branches have signatures $\sigma_0, \sigma_1 \in \{+1, -1\}$, and use the confluent basis defined by*

$$S_m = \begin{pmatrix} 1 & 1 \\ 0 & m \end{pmatrix}.$$

The pulled-back branch form is

$$J_m = (S_m^{-1})^\dagger \begin{pmatrix} \sigma_0 & 0 \\ 0 & \sigma_1 \end{pmatrix} S_m^{-1} = \begin{pmatrix} \sigma_0 & -\sigma_0/m \\ -\sigma_0/m & (\sigma_0 + \sigma_1)/m^2 \end{pmatrix}.$$

A scalar-renormalized finite nondegenerate critical limit exists if and only if the signatures are opposite. In that case,

$$\boxed{\sigma_1 = -\sigma_0 \quad \Longrightarrow \quad mJ_m \longrightarrow \begin{pmatrix} 0 & -\sigma_0 \\ -\sigma_0 & 0 \end{pmatrix}.}$$

For equal signatures, every scalar normalization that removes the m^{-2} divergence has a degenerate limit; in particular,

$$m^2 J_m \longrightarrow \begin{pmatrix} 0 & 0 \\ 0 & 2\sigma_0 \end{pmatrix}.$$

Corollary 17.15 (No finite positive critical branch observable)**.** *Every endomorphism commuting with the nonzero critical Jordan generator is $C = aI + bN$. If $C^2 = I$, then $C = \pm I$, and $GC = \pm G$ remains indefinite. Hence no bounded critical involution commuting with the dynamics converts the canonical hyperbolic form into a positive metric. The divergence of the finite-mass branch-sign involution is therefore forced by the loss of the semisimple branch decomposition, not by an unfortunate basis choice. This is a local critical-root obstruction, not a full quantum no-go theorem.*

The inverse is

$$\mathcal{T}(z, \mu)^{-1} = \begin{pmatrix} z^{-1} & [z(z-\mu)]^{-1} \\ 0 & (z-\mu)^{-1} \end{pmatrix},$$

and

$$\frac{1}{z(z-\mu)} = \frac{1}{\mu}\left(\frac{1}{z-\mu} - \frac{1}{z}\right) \longrightarrow \frac{1}{z^2}. \tag{15}$$

The critical double pole is therefore the confluent difference quotient of the two finite-mass simple poles.

Theorem 17.16 (Confluent projectors and local contour)**.** *Normalize the massive vector by*

$$u_m = u_0 + mv + O(m^2)$$

and use the confluent basis (u_0, v)*. The branch-to-confluent matrix and the two spectral projectors are*

$$S_m = \begin{pmatrix} 1 & 1 \\ 0 & m \end{pmatrix},$$

$$P_0(m) = \begin{pmatrix} 1 & -1/m \\ 0 & 0 \end{pmatrix}, \qquad P_m(m) = \begin{pmatrix} 0 & 1/m \\ 0 & 1 \end{pmatrix}.$$

Hence

$$P_0 + P_m = I, \qquad mP_0 \longrightarrow -N, \qquad mP_m \longrightarrow N, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

and $\|P_0\|, \|P_m\| \asymp |m|^{-1}$*. Equivalently, after normalizing the confluent coordinate by the spectral separation* $\delta_m = \nu_n m + O(m^2)$*, the projector scale is* $O(|\delta_m|^{-1}) = O(|\nu_n m|^{-1})$*.*

For a small contour enclosing only the two resonances, the local meromorphic evolution contribution satisfies

$$\boxed{\mathcal{U}_m(t) \longrightarrow e^{i\omega_n t}\,(I + i\nu_n t N)\,.}$$

Proof. The projector formulas are

$$P_0 = S_m \operatorname{diag}(1,0) S_m^{-1}, \qquad P_m = S_m \operatorname{diag}(0,1) S_m^{-1}.$$

Residue calculus on the local contour gives

$$\mathcal{U}_m(t) = e^{i\omega_n t} P_0(m) + e^{i\omega_m t} P_m(m) = \begin{pmatrix} e^{i\omega_n t} & \dfrac{e^{i\omega_m t} - e^{i\omega_n t}}{m} \\ 0 & e^{i\omega_m t} \end{pmatrix}.$$

Since $\omega_m - \omega_n = \nu_n m + O(m^2)$, the off-diagonal entry tends to $i\nu_n t e^{i\omega_n t}$. □

Equation (15) is the divided difference of the simple resolvent. More generally,

$$\frac{F(\omega_n + \nu_n m) - F(\omega_n)}{\nu_n m} \longrightarrow F'(\omega_n)$$

for analytic F. Taking $F(\omega) = e^{i\omega t}$ gives the local Jordan polynomial, while differentiating the massive eigenvector shows

$$V_1 \equiv \partial_m V_m|_{m=0} \pmod{\mathbb{C}V_0}.$$

Remark 17.17 (Time-domain boundary)*.* Theorem 17.16 proves the Jordan polynomial for the isolated local two-pole contour contribution. It does not supply a global retarded inverse-Laplace deformation or control the non-pole contour.

Remark 17.18 (Higher filtered jets)*.* If $k+1$ filtered branches coalesce linearly and the branch basis is replaced by its confluent divided-difference basis, the limiting connection has a chain of length $k+1$. Its local resolvent has poles through order $k+1$, and the local contour contribution is

$$e^{i\omega_n t} \sum_{j=0}^{k} \frac{(it)^j}{j!} N^j.$$

The present Bach self-extension is the first nontrivial case $k = 1$.

Theorem 17.19 (Confluent branch involution). *Let C_m act by $+1$ on u_0 and by -1 on u_m. In the confluent basis,*

$$C_m = \begin{pmatrix} 1 & -2/m \\ 0 & -1 \end{pmatrix}.$$

It has no finite critical limit, but

$$\boxed{\lim_{m\to 0} mC_m = -2N.}$$

The projectors $(I \pm C_m)/2$ have residues $\mp N$. Since $m = (i\omega/2)\tau$,

$$\boxed{\lim_{\tau\to 0} \tau C_\tau = \frac{4i}{\omega} N.}$$

Thus the unique nilpotent in the dual-number commutant is exactly the renormalized residue of the finite-mass branch-sign observable.

Proof. The equation $C_m(u_0 + mv) = -(u_0 + mv)$ gives $C_m v = -2u_0/m - v$. This proves the matrix and the stated limits. $\square$

Theorem 17.20 (Critical singularity of positive branch metrics). *For real signed $m \neq 0$, suppose a positive branch metric is*

$$H_{\mathrm{br}}(m) = \mathrm{diag}(h_0(m), h_1(m)), \qquad 0 < h_* \le h_0, h_1 \le h^*.$$

In the confluent basis,

$$H_m = (S_m^{-1})^\dagger H_{\mathrm{br}} S_m^{-1} = \begin{pmatrix} h_0 & -h_0/m \\ -h_0/m & (h_0 + h_1)/m^2 \end{pmatrix}.$$

Its condition number satisfies

$$\boxed{\mathrm{cond}(H_m) \asymp |m|^{-2}.}$$

In the spectral-separation normalization this is $O(|\nu_n m|^{-2})$. It is the physical-axis specialization of the universal gap law $\mathrm{cond}\, H \asymp |\Delta|^{-2}$ *in Theorem 17.9. No scalar rescaling of H_m has a bounded, positive-definite, nondegenerate critical limit.*

Proof. One has

$$\det H_m = \frac{h_0 h_1}{m^2}, \qquad \lambda_{\max}(H_m) = \frac{h_0 + h_1}{m^2} + O(1),$$

and therefore

$$\lambda_{\min}(H_m) \longrightarrow \frac{h_0 h_1}{h_0 + h_1} > 0.$$

Scaling slower than m^2 leaves the largest eigenvalue unbounded; scaling like m^2 gives a rank-one limit; and scaling faster gives zero. $\square$

Theorem 17.21 (Renormalized Krein limit). *Let $J_{\mathrm{br}} = \mathrm{diag}(1, -1)$. In the confluent basis,*

$$J_m = (S_m^{-1})^\dagger J_{\mathrm{br}} S_m^{-1} = \begin{pmatrix} 1 & -1/m \\ -1/m & 0 \end{pmatrix},$$

so

$$\boxed{mJ_m \longrightarrow \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.}$$

Moreover

$$H_m = J_m C_m = \begin{pmatrix} 1 & -1/m \\ -1/m & 2/m^2 \end{pmatrix} > 0, \qquad m^2 H_m \longrightarrow \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}.$$

Thus the pure-Weyl hyperbolic plane is the renormalized critical limit of the opposite-sign finite-mass branch form, whereas its positive C-majorant becomes rank-one.

Proposition 17.22 (Canonical pseudospectral scale)**.** *For the leading two-parameter normal form,*

$$T(z; m, \epsilon)^{-1} = \frac{1}{z^2 - \nu_n m z + c_n \epsilon} \begin{pmatrix} z - \nu_n m & 1 \\ -c_n \epsilon & z \end{pmatrix}.$$

Put

$$\zeta = z - \frac{\nu_n m}{2}.$$

Then

$$z^2 - \nu_n m z + c_n \epsilon = \zeta^2 - \frac{\Delta^2}{4}.$$

In the unresolved regime $|\zeta| \gg |\Delta|$, the upper-right response is

$$\boxed{(T^{-1})_{12} = \frac{1}{\zeta^2} \left[1 + O\left(\frac{\Delta^2}{\zeta^2} \right) \right].}$$

When $|\zeta| \lesssim |\Delta|$, the two simple poles must be resolved separately and their projectors have size $O(|\Delta|^{-1})$.

At the exact critical point,

$$\left\| \mathcal{T}(z, 0)^{-1} \right\| \asymp |z|^{-2}.$$

Hence the local ε-pseudospectral radius is

$$\boxed{|z| \asymp \sqrt{\varepsilon}}$$

rather than the simple-pole scale $O(\varepsilon)$. For $m \neq 0$, the crossover is controlled by the actual gap. Centering at ζ, rather than comparing raw z only with Δ, is essential near the exceptional parabola, where the common pole location can move even as the gap vanishes.

Remaining scalar/tensor projection audit

Let R reconstruct the tensor carrier from the scalar master field and let P project the tensor equation back to the scalar equation. The corresponding exact target is

$$K_{\text{radial}} - P\mathcal{A}R = LQ_0 - Q_0 L$$

in the augmented pencil, with its endpoint component recorded explicitly. Section 16 already identifies the tensor parent coefficient through the massive resolvent. Proving this commutator identity would additionally identify every scalar reconstruction normalization and endpoint term coefficientwise; it is no longer needed to establish the nonzero parent double pole.

18 Computational certificates and claim boundary

The computer-assisted inputs are separated from the exact derivations above.

Certificate family	Claim used here	Evidence type
Axial triangular preflight	Exact RW/RW/Maxwell filtration and reconstruction	Exact rational
Projective cocycle	Gauge law, nontrivial extension class, mass normal form, static cokernel	Exact rational
Critical parent mass jet	TT mass variation and difference quotient	Exact covariant/reduced
Analytic continuation	Analytic Jost germs on the declared QNM domain	Exact continuation gate
RW/Maxwell simplicity	Physical-axis simplicity and scalar endomorphisms	Exact rational
Full Evans contour	Zero-free boundary and winding +1	Validated interval
Local selector	$\delta_\tau \neq 0$ on the root disk	Validated interval
Spin-one local unit	Full three-factor Smith type $(0, 0, 2)$	Validated interval
Fredholm promotion	Finite-interval rank-one radial Green pole and transparent-cut input	Exact reduction plus validated inputs
Null-infinity reconstruction	Exact E/R metric heads, odd radiation gauge, nonzero Einstein Bondi shear, and weakened generalized falloff	Exact symbolic
Observable transfer	Analytic source/reconstruction maps and rank-one resonant overlaps	Exact conditional reduced-mode
Coherent forcing and matched drive	Isolated critical kernel and coefficient-space L^2 budget	Exact conditional reduced-mode
Quasinormal logarithmic partner	Fixed-frequency Coulomb cancellation, total null-time tangent, and Jordan law	Exact reduced-mode

The theorem statements and proofs are the mathematical authority. The machine-readable records pin exact witnesses, interval margins, source hashes, and independently executable checks. A missing input, unsupported domain, timeout, or failed mutation causes rejection rather than partial acceptance.

Explicit non-results. This paper does not establish a causal spacetime resolvent, a complete QNM expansion, a global late-time asymptotic formula, a retarded inverse-Laplace deformation, a time-domain instability, the off-resonance normalization function $h(\omega)$, an all-ℓ Bach nonsplitting theorem, a certified value of c_n for a concrete physical filtration-breaking perturbation, a numerical value of ξ_n, a threshold-uniform estimate for b/a^2, a validated multi-QNM contour sum, a validated acceleration contour or numerical acceleration, an overtone EP2 tower, a global positive-metric or quantum no-go theorem, or a quantum positivity statement. It also does not establish nonzero excitation by a specified astrophysical source, detector detectability, or parameter-estimation sensitivity. The polynomial $te^{i\omega_n t}$ is proved only for the isolated local two-pole contour contribution. The coherent-drive and pulse-train laws likewise apply only to that isolated kernel; no physical matched-source construction, invariant energy-budget identity, or nonlinear saturation calculation is claimed. The exact null-infinity reconstruction proves that the generalized constant component does *not* have standard asymptotic-flatness falloff in the declared odd radiation gauge. It does not establish finite Bondi flux for that component, a large-gauge completion that removes it, or a full Newman–Penrose reconstruction. The nonzero asymptotic observation overlap of the Einstein-shaped enhanced term is proved, but its coefficient remains conditional on the source overlap and on a global retarded contour theorem.

19 Conclusion

The axial Schwarzschild Bach equation contains a repeated Regge–Wheeler factor, but the repetition is not a rational direct sum. It is a non-split self-extension whose projective class is represented by the elementary direction $(i\omega/2)(1-2/r)$. That direction is exactly the transverse-traceless critical mass jet. The static mass-direction class is nonzero for every radiative $\ell \geq 2$, while the dipole is exact. For the explicitly reduced $\ell = 2$ Bach cocycle this proves exact threshold valuation one: the singular leading frame term is exact, but the surviving linear class is not. The equivalence gauge has forced linear growth at infinity, but this is the derivative of the moving massive phase rather than an obstruction: the Coulomb tail cancels its apparent logarithmic shear. At the certified damped mode, the scalar zero is simple while the extension selector is nonzero. The resulting defect has concrete differential content: the generalized root has a nonzero Ricci-carrier quotient, the compactly cut-off exterior outgoing Green operator has a genuine rank-one double pole, its full two-layer coefficient is nilpotent, and the massive spin-two QNM moves with nonzero critical slope. The generalized state is therefore the asymptotically flat quasinormal analogue of critical gravity's logarithmic partner: modulo the Einstein mode it is the mass derivative of the massive QNM. Unlike the familiar anti-de Sitter examples, its fixed-frequency Jost derivative contains no Coulomb logarithm. Along the moving QNM branch the total derivative is organized by the phase-adapted null time $u_\sigma = t + \sigma r_*$, and the normalized Jordan chain evolves as $e^{i\omega_n t}(V_1 + itV_0)$. The exact axial metric reconstruction settles the falloff question: the Einstein root has nonzero Bondi shear, but the time-independent generalized carrier has $O(1)$, rather than $O(r^{-1})$, strain in standard odd radiation gauge. The enhanced uV_0 component remains ordinary outgoing radiation and has an explicit u/r coefficient. At the covariant level, the metric response is the mass derivative of the massive spin-two resolvent: QNM motion controls its double coefficient, while projector motion controls its accompanying simple coefficient. The second derivative of the massive QNM is fixed by the ordinary finite Regge–Wheeler resolvent and the augmented resonant pairing, furnishing the first finite correction to the confluent projector and contour limits. The ratio b_{B}/a is the corresponding spectral-velocity generating function: its residues are the selectors and its contour integral sums their infinitesimal QNM motion. At every simple nonzero QNM, nonzero first velocity is equivalent to the defective Smith branch; vanishing velocity gives the semisimple branch, while projector motion may still leave a simple pole. For higher mass jets, the exact pole hierarchy is controlled by the order of spectral contact rather than merely by the number of derivatives. The canonical logarithmic forms Θ_1 and Θ_2 sharpen this to local spectral kinematics: their residues are the QNM velocity and acceleration, and their weighted contour moments sum both quantities over analytic QNM cells. The second critical jet then separates squared velocity, acceleration, and projector motion among its triple, double, and simple coefficients. For the isolated certified damped mode, the Jordan envelope peaks at $t = 1/\operatorname{Im}\omega_n$ and subsequently decays. Any analytic chain of source insertion, reconstruction, and observation carries the leading rank-one pole by its resonant values alone. The critical pair therefore has the exact real repeated-root template

$$e^{-\gamma t}\big[(a_0 + a_1 t)\cos\Omega t + (b_0 + b_1 t)\sin\Omega t\big],$$

provided the source and observation overlaps are nonzero. The associated divided-difference basis remains finite across the critical confluence and separates the Jordan-like, crossover, and resolved regimes by $|\delta|/\gamma$. The isolated polynomial component has finite integrated amplitude and derivative norm. Proving the two overlaps for a specific null-infinity observable and physical source, and embedding the local contour in the global causal evolution, remain separate tasks. Within the same local boundary, phase-locked forcing integrates the critical kernel once more: its early response is quadratic rather than linear, and its steady amplitude has a squared Lorentzian denominator. Finite phase-matched pulse trains exhibit the corresponding damped arithmetic–geometric sum. For a fixed coefficient-space L^2 budget, the unique optimal temporal profile is the time-reversed conjugate critical kernel and its attainable amplitude saturates at $|C|\sqrt{E}/(2\gamma^{3/2})$. The low quality factor of the certified mode confines useful coherent preparation to one or a few damping times.

The physical mass deformation is filtration-preserving rather than a generic square-root unfolding. In the two-parameter phase diagram, the protected Einstein branch stays fixed and the mass axis splits linearly, whereas reverse Einstein-to-carrier mixing produces square-root branching and resonance exchange. The actual spectral gap controls projector growth and positive-metric conditioning, and recovering the physical slope requires filtration-breaking error $o(m^2)$. The gap-renormalized projector difference converges path-independently to the unique nilpotent commutant generator. Along the physical axis, the local two-pole contour converges to the Jordan polynomial and the renormalized opposite-sign branch form becomes the pure-Weyl hyperbolic plane. A global causal time-domain interpretation still requires the separate retarded contour theory stated above. On the critical root fiber these facts are one geometry: a nonzero self-adjoint length-two nilpotent forces the nondegenerate compatible form to be hyperbolic, makes the Einstein root null, and realizes the full double-pole coefficient as a null rank-one square-zero map. A finite nondegenerate first-order confluence of branch forms exists precisely for opposite signatures; no positive-definite form or bounded commuting involution survives the critical Jordan limit.

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